# Character Sentences as Quantitative Metrics: Leveraging Large Language Models to Measure Literary Characterization

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## Abstract

This thesis investigates literary characterization through computational literary studies by introducing a novel analytical unit termed "Character Sentences" (CS)—units explicitly providing descriptive or action-related information about literary characters. Three primary contributions are presented in the thesis: (1) a narratological definition and criteria of character sentences; (2) two gold-standard datasets, HPCS (*Harry Potter Character Sentence_clause-level* & *Harry Potter Character Sentence_full-sentence*), meticulously annotated from the *Harry Potter* series, designed to benchmark automated character sentence extraction tasks; and (3) a natural language processing (NLP) pipeline integrating large language models (LLMs) to automatically identify character sentences and accurately attribute them to corresponding characters, suitable for texts of any length. The gold-standard datasets demonstrate high inter-annotator agreement, with Krippendorff's α values exceeding 0.80 ($\alpha_{\text{HPCS\_clause-level}} = 0.81$; $\alpha_{\text{HPCS\_full-sentence}} = 0.86$). The proposed NLP pipeline comprises four modules: (1) a text cleaning module; (2) a sentence segmentation module aligned with the character sentence definition; (3) a zero-shot LLM processing module employing the LangGPT prompting framework and two-stage coreference resolution reasoning; and (4) a dependency parsing-based filter module enhancing the accuracy of character attribution. Empirical evaluations indicate the pipeline achieves a robust performance, yielding an F1 score of 94.51% in character sentence identification and an accuracy of 84.88% in character attribution on the HPCS_full-sentence dataset. This research is the first to explicitly define character sentences and develop an automated, theory-informed, sentence-level approach integrated with LLMs for character sentence extraction. It addresses critical gaps in computational literary studies and underscores the efficacy of LLMs and prompt engineering within literary analysis. The datasets and source code developed in this thesis are

publicly accessible on GitHub to facilitate further research and methodological advancements in the field.

## Lay Summary

This research introduces a new, quantifiable method called "Character Sentence" (CS) for analyzing how literary texts portray characters. A CS is defined as any sentence or dialogue providing descriptive or action-based details about a character, helping readers form impressions of that character. The study (1) establishes clear, theory-driven criteria for identifying CS, (2) creates annotated Gold datasets to benchmark automated CS retrieval, and (3) develops a computer-based pipeline using advanced language models to identify CS and attribute them to the portrayed characters. The research demonstrates that both humans and artificial intelligence can consistently recognize these characterization sentences, bridging traditional literary analysis and modern computational methods. By openly sharing datasets and tools on GitHub, this work supports future studies aiming to explore characterization in literature more efficiently and systematically, enhancing our understanding of how stories communicate complex character information.

## Preface

This thesis presents original research conducted by the author. The identification and design of the research program, including the formulation of research objectives, defining the concept of Character Sentence (CS), and designing the methodological framework, were independently undertaken by the author, with guidance from the supervisory committee. The author collaborated with another person (who would like to remain anonymous) in the creation of annotated datasets, and completed the development of the natural language processing pipeline, computational experiments, and all analyses independently. The author independently analyzed and interpreted the research data, drawing conclusions that contribute to the fields of cognitive narratology and computational literary studies. The datasets and computational tools developed in this project have been openly shared on GitHub to facilitate further scholarly investigation.

Generative artificial intelligence (AI) is utilized in this research as a main component of the analytical pipeline of this research. Large language model Gemini 2.5 Pro is integrated into the Ext-cs character extraction pipeline designed by this study. Additionally, this thesis uses generative AI to polish and proofread the manuscript for readability.

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## List of Abbreviations

- CLS: Computational literary studies
- CS: Character sentence
- LLM: Large language model
- NLP: Natural language processing

## Acknowledgements

I remember, in my third undergraduate year, I found an abstract about modeling the personalities of characters in Eileen Chang's novel *The Golden Cangue*. This introduced me to narratology, digital humanities, and computational literary studies, reshaping my view of beloved literary characters. That's where everything began.

I remember my freshman Python class, where my final project was a text editor named Xiaobo, after my favorite Chinese writer Wang Xiaobo, who was also a geek and created his own text editor with C++ in the early 90s. When asked why I created it despite existing software, I simply replied it was customized for me, though I wanted to say it was a tribute. No one in that computer science classroom seemed to care about Wang Xiaobo, much less literature.

I remember in my first year at UBCO, when discussing Eileen Chang and Wang Xiaobo in a creative writing class, I sensed confusion among my fellow young writers who tried but struggled to relate. It is a metaphor—here, my experiences, memories, and feelings from my homeland often seemed meaningless. Research, academic knowledge, and scholarly exchanges became my refuge.

Even my research, digital humanities and literary characterization, felt marginal in the drowning land of humanities. Finding peers with similar interests at UBC proved nearly impossible, and brief connections made at academic events soon faded. At times, even I questioned the purpose of my work.

Yet there were meaningful moments. Each research breakthrough brought me closer to the literary figures I'd befriended since childhood. I treasured building data and scripts, gaining insights from my supervisory committee, peers, and even ChatGPT.

My heartfelt gratitude goes to my supervisor, Kyong, whose unwavering support and kindness helped me navigate academic and personal challenges abroad. My co-supervisor, Liane, provided profound insights into research itself and inspired me through her passion and integrity. Emily, my committee member and PI at ReMedia Lab, continually offered fresh perspectives and essential guidance in literary studies and digital humanities. Her spirit is my parachute to land in academia. Fatemeh, my other committee member, provided practical insights from software engineering that strengthened my confidence from the outset.

Without my committee, friends, and loved ones here and back home, this work wouldn't exist. Thank you to everyone who accepted, supported, and believed in me.

I still remember tracking down the full article of the abstract on Eileen Chang's characters' personalities. The corresponding author told me that the first author graduated and left academia; their work never went beyond an abstract. That moment of pity deeply caught me. I remember when I began the journey of grad school, amidst the broken heart of the post-COVID world, leaving home, and confusion about the future, I told myself, *Your only goal here is to create something clear and rigorous—something another curious soul might one day find meaningful.* That would be enough.

I remember. And with all the love and support, I did it.

## Dedication

To all the authors of the books that I have read; to all the fellow writers who believe in the magic of literature. Keep working on it.

To my mother and father, who have, and had, always encouraged me to write and create. I love you dearly.

To Guijing and Qingshui, the two lovely characters of my first sci-fi novel. You are two pieces of shining gold.

# Chapter 1: Introduction

We read, and we fall in love with the people on the page. But how do figures wrought from nothing but ink and paper acquire a palpable life? Characterization is the literary process of the transformation from words to characters. *Cognitive narratology theory* asserts that readers distil attributes from the text and, block by block, assemble them into living characters in their minds. How does this happen? At what textual granularity does this interpretive understanding occur? This thesis addresses these questions through the lens of computational literary studies.

The thesis proposes a novel analytical unit for literary characterization at a large scale, the *Character Sentence* (CS), which is defined as a sentence that supplies information that enables readers to ascribe attributes to a character. Building on this concept, I designed a natural language processing (NLP) pipeline that integrates large language models (LLMs) to automatically extract character sentences from literary works of any length. The extracted character sentences can serve as the substrate for downstream analyses, such as emotion profiling, agency and action patterns, and social-network construction, offering a quantitative path to modelling characterization at scale.

## 1.1 Background and Motivation

Characters are central to fiction. While early narrative theorists treated characterization as a property inscribed in the text itself, their post-classical counterparts re-cast it as an activity completed in the reader's mind. Early theories were resolutely text-oriented: E.M. Forster's (1927) famous opposition between "flat" and "round" figures indexed complexity on the page, and structuralist narratologists, such as Roland Barthes and Shlomith Rimmon-Kenan, "pictured character as mere words or a paradigm of traits described by words" (Jannidis, 2012, p.3).

Post-classical approaches, by contrast, relocate agency to reception. *Rhetorical narratology* conceives characters as strategic prompts that secure readers' "strong desire for the success or failure of those we love or hate" (Booth, 1983, p. 125; Phelan, 1989). Later, the rhetorical view of characters turned into the narrative ethics—the moral value of stories, storytelling, and characters (Phelan, 2013). Cognitive narratology refines this insight, framing characters as mental models assembled via Theory-of-Mind inference (Palmer, 2004; Zunshine, 2006). Finally, *psychoanalytic criticism*—drawing on Freud and Jung—reads fictional figures as case studies in human motivation, exemplified by analyses of Dostoevsky's *Crime and Punishment* or Woolf's *Mrs. Dalloway* (Ogden, 2013).

These theoretical developments share an interdisciplinary impulse: characterization is now viewed as a triadic interaction among author, text, and reader, informed by psychology and cognitive science. Correspondingly, research methods have expanded beyond traditional close reading to embrace approaches from other fields, such as computer science.

Notably, the term "reader" within literary characterization, and specifically within this research, refers to the "implied reader"—an imagined audience embedded within the text rather than actual, flesh-and-blood readers. Originally proposed by Wayne C. Booth (1983) as the author's envisioned recipient, the concept was further articulated by Wolfgang Iser (1994, 1995), who defined the implied reader as a textual structure anticipating the presence of an actual reader and guiding their interpretation. For Iser, the implied reader represents a dynamic, meaning-making process wherein the text presents "schematized views," gaps, or indeterminacies, and the reader (occupying the implied reader's position) fills these gaps to concretize the literary work (Prince, 2013). In studies of literary characterization, conclusions regarding reader-text interactions are not derived from empirical observation or interviews with real readers; instead, they involve theorizing the role and functions of the implied reader constructed by the text itself.

*Computational literary studies* (CLS) apply natural language processing (NLP) and machine learning to large corpora, enabling a "distant reading" (Moretti, 2000) of both the "great unread" text ignored by literary history (Cohen, 2002) and born-digital genres such as online novels and fanfiction. CLS has addressed semantic change, stylometry, narrative structure, sentiment, and—most relevant here—character analysis (Bories et al., 2022; Kukkonen and Nielsen, 2018; Piper, 2018). Within character-focused CLS, researchers have mapped social networks (Labatut and Bost, 2020; Kubis, 2021), inferred latent personas (Bamman et al., 2013, 2014), tracked sentiment and emotion arcs (Jacobs, 2019; Tayal et al., 2022), modelled relationship dynamics (Jayakumar et al., 2022), and generated character profiles (Papoudakis et al., 2024). These research topics rely heavily on NLP tasks such as named-entity recognition (NER), coreference resolution, event detection, and sentiment analysis, yet they typically report results either at the token/mention level or at coarse plot segments; sentence-level evidence remains sparsely exploited. Moreover, most implementations optimize for NLP benchmarks rather than for narratological questions, leaving theory-driven character modelling underdeveloped.

Large language models (LLMs) have recently extended the CLS toolkit, improving text classification, automatic annotation, sentiment mining, coreference resolution, and quotation attribution (Carroll, 2024; Hicke and Mimno, 2024; Martens and De Greve, 2023; Michel et al., 2025). LLMs already yield nuanced literary insights at the corpus scale. However, LLM-based studies rarely target characterization itself, and their capacity to isolate, describe, or track fictional personas is largely untested.

This thesis addresses these gaps by introducing **Character Sentences**—sentence-level units that convey a character's traits, emotions, perceptions, or actions. It develops an end-to-end NLP pipeline that integrates state-of-the-art large language models with rule-based filters to identify and attribute these sentences. This design demonstrates how an existing LLM can be harnessed for theory-driven character analysis while remaining fully transparent and reproducible. The pipeline is envisioned to benefit literary scholars in understanding the literary yet cognitive process of literary characterization; scholars can feed the output into downstream tasks such as emotion classification and personality profiling for finer-grained interpretation of literary characters.

## 1.2 Research Rationale

This project is driven by four intertwined needs—conceptual, granular, methodological, and empirical—that together motivate a new approach to computational studies of literary characterization.

1. **Conceptual rationale:** To develop a theory-informed approach toward the computational studies of characterization.

Most CLS work stops at surface-level recognition of named entities or social ties, instead of cues that the readers rely on to infer character attributes, such as actions, mental states, and dialogue (Jaipersaud et al., 2024). Capturing these cues yields a fuller portrait of character agency and reveals how authors guide readers' Theory-of-Mind construction of fictional figures. Building on cognitive narratology, this thesis therefore introduces the character sentence: any clause or sentence likely to update a reader's mental model of a character. By operationalizing this notion in an NLP pipeline that detects and attributes such sentences, the study provides an empirical bridge between narrative theory and large-scale text analysis.

2. **Granularity rationale:** To promote sentence-level units targeting portrayal cues.

Prior pipelines aggregate signals at the token or scene level, or include every sentence that merely *mentions* a character; this method does not guarantee that the analyzed sentences are centered to the portrayal of a character, producing noisy results (Piper, 2024). Because the sentence is fiction's primary syntactic and semantic unit (Ohmann, 1966), restricting analysis to portrayal-relevant sentences offers a sharper lens: it preserves the local context necessary for interpreting actions and mental states while remaining short enough to map narrative pacing, emotional arcs, and distribution of character information with precision. The result is cleaner input for downstream tasks such as sentiment or personality modelling.

3. **Methodological rationale:** To utilize LLM prompting in characterization tasks.

Widely used toolkits like BookNLP and FanfictionNLP rely on Bidirectional Encoder Representations from Transformers (BERT)-scale encoders (≈110–340 M parameters) and handcrafted rules, constrained to 512-token windows and slow when applied to book-length texts. This thesis integrates Gemini 2.5 Pro, a multi-billion-parameter LLM with a 1 M-token context, into an end-to-end pipeline for automatic character-sentence detection and attribution. Prompt-based LLMs reduce feature engineering overhead, accommodate long passages (crucial for free indirect discourse), and expose intermediate reasoning that can be inspected by humanists, thereby enhancing both scalability and interpretability.

4. **Data/evaluation rationale:** To create benchmark datasets for sentence-level characterization.

The field lacks a public dataset that labels sentences by their contribution to characterization, making it impossible to compare methods, reproduce results, or train supervised models. This thesis releases the first gold-standard benchmark—annotated excerpts from the *Harry Potter* series—together with clear evaluation protocols. The resource establishes a common yardstick, accelerates methodological innovation, and lowers the barrier to entry for future CLS research on character analysis.

### 1.3 Research Objectives

The aim of this research is to establish a cognitive narratology-informed, large language model (LLM)-integrated framework for quantitative analysis of literary characterization, centered on the proposed analytical unit of the **character sentence**. The research objectives are:

1. **Conceptualization:** Define and formalize the notion of a Character Sentence concerning the narratological accounts of characterization.
2. **Pipeline design:** Develop a character sentence extraction pipeline and Gold benchmarking datasets.
3. **Evaluation:** Optimize model selection and prompt design; conduct three controlled experiments to measure precision, recall, and ablation effects.
4. **Open science:** Release all datasets, code, and prompts to facilitate reproducible research in CLS.

### 1.4 Research Questions

The thesis addresses four research questions:

**RQ1 (Conceptual)** How can a Character Sentence be defined such that it is both distinctive in narratology and computationally tractable?

**RQ2 (Methodological)** To what extent can an API-accessible LLM accurately detect and attribute Character Sentences in complex literary prose?

**RQ3 (Methodological)** What are the strengths and limitations of integrating rule-based NLP and LLM reasoning for sentence-level characterization?

**RQ4 (Engineering)** Which prompt-engineering strategies maximize LLM performance in this extraction task?

These questions correspond directly to Objectives 1-4 above, and provide a structured framework for the experiments and discussion that follow.

## 1.5 Methodological Overview

To develop a theory-informed approach toward computational character studies and develop a tool to retrieve accurate sentence-level character portrayal cues, this thesis follows a theory-to-tool design-science workflow: a process that translates theoretical ideas into practical methods or toolkits to solve problems (Romme and Holmström, 2023). Building on cognitive narratology theories, a narratological concept—the *Character Sentence* (CS)—is formalized and operationalized, and instantiated in an end-to-end extraction pipeline. The pipeline is evaluated on a purpose-built Gold corpus.

### 1.5.1 Conceptualization

- **Conceptual definition.** Building on narratology theories that emphasize textual cues and the reader's interpretation of characterization, a CS is defined as any sentence that enables readers to infer a stable attribute, emotion, or motive of a fictional character.
- **Operational definition.** The definition is translated into four reproducible rules to derive the annotation scheme and algorithm design (details in §3.2).
  - **Speaker presence.** To identify quotes and dialogues and attribute them to the speaking character(s).
  - **Attribute predicate.** To identify narrative sentences with descriptions or actions that update the readers' (implied reader) mental model of the character.
  - **Grammatical subjecthood.** To attribute narrative sentences to the acting character(s).
  - **Reference scope.** To expand coreferences (she, he, them) in the sentences for attribution.

### 1.5.2 Gold-standard Data

A Gold-standard dataset or a Gold dataset is a meticulously curated, high-quality dataset used as a benchmark to evaluate the performance of machine learning models or computational methods. This thesis aims to build and publish Gold datasets for the automated CS extraction task with two main steps: corpus sampling and data annotation.

- **Corpus sampling.** A subset of the *Harry Potter* novels is selected: a 200-sentence continuous passage sampled from each book, covering 41 named characters in total. The average word count for each passage is 3037 words; the average token count for each passage is 4032.57 tokens (with OpenAI's tokenizer, see https://platform.openai.com/tokenizer). The sampled passages are believed to be of a reasonable length for both human annotators and LLMs' context windows, and both main and secondary characters are included in the corpus.

- **Annotation protocol.** Two annotators label CS boundaries, attributions (portrayed characters of the sentence), and sentence type (narration vs. direct speech). Krippendorff $\alpha > 0.80$ confirms reliability.

### 1.5.3 Pipeline Architecture

1. **Text normalization.** Whitespace and Unicode cleanup.
2. **Senticizer.** Quotation-aware splitter that inserts sentence IDs.
3. **LLM core.** Gemini 2.5 Pro processes 1,500-token windows via a LangGPT prompting framework with two-stage coreference reasoning, returning candidate CSs and character lists.
4. **Dependency filter.** spaCy parses each candidate CS text; only characters labelled as *nsubj*/*nsubjpass* (grammatical subject as an action/passive action) are retained.

### 1.5.4 Evaluation & Optimization

- **Model comparison.** Four state-of-the-art LLMs, LLaMA-3-8b, Mistral-7b, GPT-4o, and Gemini 2.5 Pro, are tested on a small sample to select the optimized model for the pipeline.
- **Quality metrics.** Precision, recall, and F1 on (i) CS identification, (ii) character attribution, and (iii) sentence-type labeling.
- **Prompt ablation**. Chain-of-thought, LangGPT template, and coreference expansion are toggled to measure incremental gains.

### 1.5.5 Outputs

All code, prompts, and the HPCS Gold datasets are released under the MIT License (https://github.com/OdessyLiu/Ext-CS), enabling future implementations and improvements.

## 1.6 Significance

The principal literary-theoretical contribution of this thesis is to demonstrate that the textual cues of characterization can be detected systematically. Using large language models to emulate human reading, the study operationalizes narratological insight through character sentences. Following Booth's (1983), Margolin's (1983), and modern cognitive narratology's view of characters as vehicles of author–reader communication, the character-sentence framework renders that process empirically visible at the sentence scale: scholars can now trace how authors project a persona and how readers infer it, sentence by sentence, across entire corpora. The same framework also offers creative writers a diagnostic tool, revealing what effective characterization looks like at the sentence level, and providing a concrete basis for writing instruction and assessment.

Computationally, this thesis addresses the "distant reading" method proposed by Moretti (2000). This study couples NLP with large language models to identify character sentences across complete novels, extending characterization research from a handful of texts to thousands. The resulting quantitative evidence complements, rather than replaces, qualitative interpretation by making claims measurable and reproducible; any researcher who applies the released code and data can replicate the results, test narratological hypotheses, and uncover large-scale patterns of portrayal.

The designed character sentence extraction pipeline has the following features: it extracts character sentences in a single pass from raw text, integrates Gemini 2.5 Pro prompting, is evaluated on the first publicly available Gold dataset for characterization, and is released under an open license. The pipeline closes the gaps in previous CLS character analysis tools that are not open-sourced, not characterization-centered, not API-accessible, and lacking Gold standard benchmarks. The openly released pipeline and Gold dataset also answer to the calls in digital humanities for reproducible, explainable, and open methods and tools (Ries et al., 2024).

Fine-grained extraction improves every subsequent analysis. For example, Tayal et al. (2022) traced the emotional trajectory of *Harry Potter* characters by including every sentence that merely mentions their names, in which references and genuine portrayals were mixed together, adding noise to the curve. By contrast, a character-sentence approach filters in only those sentences where the character is the grammatical subject of an action, feeling, or thought, yielding far cleaner—and therefore more reliable—sentiment arcs. The precision in portrayal-related sentence extraction also has industrial applications: AI-narrated audiobooks could adjust the narrating tone and pacing by detecting the emotional polarity and arousal encoded in character sentences, facilitating a narration that is more lively and human-like (Pan et al. 2021).

In sum, the thesis delivers a theory-informed, reproducible pipeline that expands the scale, precision, and applicability of characterization research, providing both a scholarly foundation and a practical tool for future literary and industrial applications.

## 1.7 Thesis Structure

The remainder of the thesis implements this theory-to-tool program in four chapters, moving from literature review to definition, pipeline design, and synthesis.

Chapter 2, the Literature Review, surveys three strands of prior work: narratological theories of characterization, computational approaches to characters in literary corpora, and emerging applications of large language models in the digital humanities.

Chapter 3, Definition of Character Sentences, develops the conceptual and operational definitions of a character sentence that underpin the study. It introduces the two-category framework of actions and descriptions, formalizes identification rules, and derives the annotation scheme used later for the Gold standard data.

Chapter 4, Character Sentence Detection and Attribution Pipeline, presents the methodological core of the thesis. It describes the construction of two gold datasets from the *Harry Potter* corpus, details the four-module pipeline (cleaning, sentence segmentation, LLM processing, dependency filter), and reports two experiments that evaluate pipeline quality and optimize prompt design.

Chapter 5, Conclusion, synthesizes the main findings, reflects on methodological limits, and outlines avenues for future research and application.

# Chapter 2: Literature Review

This chapter reviews the scholarly literature relevant to the two central dimensions of this study: the theoretical foundations of literary characters and characterization, and the methodological developments in computational literary studies on characters. This review situates the present research within the broader fields of literary scholarship, digital humanities, and computational literary studies (CLS).

§2.1 reviews the narrative theories of literary characters and characterization, covering classical structuralist narratology, psychological approaches, and cognitive narratology. §2.2 introduces computational literary studies on characters, beginning with a brief overview of CLS, and proceeding to examine the core research tasks, granularity and methods, existing datasets, and the recent incorporation of large language models (LLMs). Together, these sections establish the conceptual and methodological framework for the pipeline proposed in this thesis. §2.3 synthesizes previous research and addresses the positioning of the current study.

## 2.1 Narrative Theories of Literary Characters and Characterization

Before addressing how characterization occurs, it is essential to clarify what a literary character actually is. Early narrative theories can be classified into three primary perspectives: character as textual entity, character as narrative function, and character as person. Classical narratology, particularly the structuralists, encompasses the first two perspectives. The psychoanalytic or mimetic approach, in contrast, regards characters as moral and psychological entities analogous to real people. Later, the readers' perception is introduced into theories of character and characterization through cognitive narratology and reader response theory.

### 2.1.1 Classic Views Toward Characters

Even before structuralist narratology became dominant, philosophers, authors, and literary critics had diverse approaches to literary characters. Aristotle, in his *Poetics*, famously states: “Without action there cannot be a tragedy; without characters there can” (Murray, 1916). This statement highlights Aristotle’s “plot-over-character” orientation, although some scholars argue that Aristotle’s concept of *mythos* differs fundamentally from the English notion of plot (Belfiore, 2000). Aristotle also described literature as the *mimesis* (imitation) of *praxis* (action), with characters being the agents of this imitation (Belfiore, 1983), thereby laying the foundation for the mimetic understanding of characters. E.M. Forster (1927) introduced the widely accepted distinction between flat and round characters: a flat character is a static character who does not change or grow, lacking depth and personality, while a round character is multi-dimensional, actively evolving across the story. Nevertheless, Forster’s distinction has been criticized for its subjectivity and lack of consistent criteria (Lanone, 2008). Early definitions and summaries of characters have paved the path for the future theorizing of characterization; in this research, action is an important evidence of characterization when identifying character sentences.

Some structuralist narratologists regard characters primarily as textual entities. Wellek and Warren (1949) assert that characters consist merely of words. Roland Barthes, in his influential work *S/Z*, describes characters as networks of *semes* or *signifiers* attached to proper names, producing a “character-effect” rather than a representation of a real person (Barthes, 1980). Similarly, Juri Lotman (1977) characterizes literary figures as sums of binary oppositions within a textual paradigm. Margolin (1989) clarifies that this textual approach treats characters as entities created through cohesive reference by proper nouns, noun phrases, and pronouns, focusing purely on intra-textual issues such as identification, substitution, reference, and cohesion. These theories are reminiscent of the fundamental task of identifying character presence in computational character analysis, such as named entity recognition (NER) and coreference resolution.

In another structuralist perspective, associated with Propp and Greimas, characters function chiefly as narrative roles. Propp (1968) identifies characters according to their "spheres of action" and narrative roles, categorizing them into seven archetypes: villain, donor, helper, princess, dispatcher, hero, and false hero. Expanding Propp's ideas, Greimas's actantial model simplifies all characters into six roles—subject, object, helper, opponent, sender, and receiver—whose interactions drive the narrative action (Greimas et al., 1983). These abstract roles gain social specificity through "semantic concretization," such as transforming a helper into a courtier or wizard, yet the underlying functional role remains intact (Margolin, 1989). Both scholars emphasize action over interiority, echoing Aristotle's claim that narrative imitates action, with characters as agents who perform these actions, and character traits are primarily those motivating visible deeds (De Temmerman and Van Emde Boas, 2017).

In summary, Margolin (1989) synthesizes six structuralist approaches to characters: as topic entities, narrative devices, textual speakers, thematic constructs, actants, or non-actual individuals. Despite the theoretical contributions of the structuralists to characterization, the structuralist's approach is criticized for the failure to differentiate clearly among multiple meanings of "character" and the lack of a coherent theoretical framework (Margolin, 1989).

The mimetic approaches treat literary characters as imagined humans whose behaviors, thoughts, and development can be understood through psychological theories. This perspective emphasizes the characters' motivation, inner conflict, personality, and development. In Freud's psychoanalytic literary criticism, for example, narratives and characters are interpreted as symbolic reflections of the author's repressed desires and unresolved conflicts; hence, the characters are viewed as exhibiting psychological conflicts and issues similar to real individuals (Ogden, 2013). Freud's (1997) analysis of *Hamlet* as analogous to *Oedipus Rex* exemplifies this interpretation of characters. Jung's archetypal criticism, meanwhile, treats characters as symbolic types from the collective unconscious, such as mother, companion, and bride (Frye, 1951). However, psychoanalytic criticism has been criticized for extending far beyond textual evidence, sometimes misapplying Freudian concepts (Brooks, 1987; Crews, 2003).

The issue with the mimetic approach of analyzing characters strictly as real people is evident: can literary texts reliably provide sufficient and accurate information to support such an analysis? Wayne Booth's concept of the unreliable narrator addresses precisely this issue, suggesting that narrators may, intentionally or not, mislead readers through lies, misrepresentations, or withheld information (Booth, 1983). Such narrative techniques can significantly shape readers' engagement with characters and plots (Kermode, 1974). Conversely, as for the structuralist's perspective, treating characters solely as textual constructs or narrative functions neglects the reader's emotional involvement and attachment to characters.

In this research, characters are not viewed as mere textual entities; characters are viewed as human-like figures that can be modeled in emotions, personalities, and consistent behavioral patterns. However, they are also not the same as equivalents to human beings, as in the mimetic approach, considering the unreliable narrator and the reader's engagement in characterization. Thus, this thesis explored beyond the classic views of literary characters to cognitive narratology and the reader response theory.

### 2.1.2 Cognitive Narratology

The role of the reader thus emerges prominently in theories of characterization. In the latter half of the twentieth century, theorists such as Booth and James Phelan emphasized the relationship between narrative technique, character portrayal, and reader response. Booth (1983) argues that narrative is inherently rhetorical; authors shape narratives deliberately to evoke specific reader responses. He introduces the concept of the implied author and the implied reader: the narrative voice distinct from the real author and the imagined reader in the position of the audience of the narration. Chatman (1980) further develops Booth's idea, proposing a communication structure that includes real and implied authors and readers:

*Real author → [Implied author → (Narrator) → (Narratee) → Implied reader] → Real reader*

Chatman's model of narrative communication captures the triadic interaction between author, text, and reader, in which the "reader" notably refers to the *implied reader*, rather than an actual human reader. The concept of the implied reader was further articulated by Wolfgang Iser (1994, 1995), who emphasized the interaction between text and reader: the text provides schematic or incomplete aspects of meaning, while the real reader actively concretizes that meaning; the implied reader thus functions as a textual construct situated between the text and real readers, structuring and guiding the interpretive process and consolidating the art of reading (Prince, 2013). In the context of this research, "reader" specifically denotes the implied reader, as no empirical studies or interviews with human readers have been conducted. Instead, the implied reader is understood as a textual role embedded within the narrative, designed to guide actual readers' meaning-making processes.

Similarly, Phelan (1989) conceptualizes characters through three interrelated aspects: synthetic (the artificial textual construction), mimetic (the lifelike aspect), and thematic (the character's role in conveying ideas). Readers respond simultaneously to characters on all these levels, integrating structuralist and psychological approaches and underscoring reader involvement. These narrative theories are fundamental to the upcoming cognitive narratology.

Since the 1990s, cognitive narratology has introduced cognitive science perspectives into narrative analysis. Herman (2009) highlights the dual focus of cognitive narratology on the minds represented in stories and the minds of readers who construct these representations. Characters are mental models assembled by readers from textual cues using cognitive processes similar to those used to understand real people (Margolin, 2007). A core concept in this approach is Theory of Mind (ToM), the ability to attribute mental states to others. Zunshine (2006) theorizes that reading fiction engages—and may, at least temporarily, exercise—this capacity. Later experiments by Kidd and Castano (2013) proved that reading fiction enhanced participants' ToM task performances. Palmer (2004, 2011) focuses on how narratives construct the ongoing, socially embedded consciousness of characters. Margolin (1983) explicitly connects characterization to readers attributing traits to human-like narrative agents, a process rooted in ToM.

Empirical evidence from psychology and neuroscience converges with cognitive narratology's claims. Functional MRI studies indicate that literary reading activates brain regions associated with social cognition and mental-state reasoning (Tamir et al., 2016). Jacobs and Willems (2018) integrate these findings into a neurocognitive-poetics model, and Keen's (2006) work on narrative empathy correlates textual strategies with measurable neurological responses (Best, 2020). These empirical studies demonstrate consistent involvement of ToM-related systems whenever readers track fictional characters. The studies have also turned cognitive narratology's, or the long history of narrative theory's, attention from the implied readers to the actual human readers. The convergence of theoretical and empirical conclusions in cognitive narratology demonstrates that the implied reader and actual readers share substantial cognitive commonalities, validating the implied reader concept as a useful theoretical construct grounded in genuine psychological processes.

### 2.1.3 Reader-Response Theory

Reader-response theory is a literary-critical approach that shifts analytical focus from the author or text to the active role of the reader in creating meaning. It asserts that literary meaning emerges primarily from the reader's interpretive engagement with the text, rather than solely from authorial intention (Cahill, 1996).

Historically, reader-response theory can be traced to I. A. Richards's (1929) early investigation into readers' emotional responses. Louise Rosenblatt subsequently emphasized authentic student interpretations over imposed pedagogical interpretations in the 1930s, formalizing her transactional theory in 1978, which argues that meaning arises from a dynamic interaction between the reader's personal experiences and the text's potentials (Kunjanman & Aziz, 2021). Roland Barthes (1968), in "The Death of the Author," similarly highlighted reader autonomy and interpretative freedom.

Various approaches within reader-response theory further illuminate reader engagement. Stanley Fish's (1970) affective stylistics views texts as existing solely through the act of reading, emphasizing how stylistic elements evoke emotional responses in readers. Norman Holland's (1989) psychological reader-response theory suggests readers unconsciously project their personal identity themes—psychological patterns or fantasies derived from lived experiences—onto narratives, employing defense mechanisms to maintain psychological equilibrium. David

Bleich's (1975) subjective reader-response theory radically denies textual objectivity, asserting that the literary text consists entirely of readers' subjective responses. Reception theory, advanced by German scholars Hans Robert Jauss and Wolfgang Iser, adopts historical and phenomenological perspectives. Jauss (1974) introduces the concept of readers' historically and culturally conditioned "horizons of expectations," while Iser (1995) theorizes reader-text interactions through the implied reader, a textual construct guiding interpretive engagement.

Reader-response theory also significantly influences literary pedagogy, emphasizing learner-centered responses to literature. Empirical research in educational settings has demonstrated that reader-response approaches enhance students' personal engagement, reflective thinking, creativity, self-directed learning, higher-order reasoning, and interpretive skills (Kunjanman & Aziz, 2021).

Compared with cognitive narratology, reader-response theory prioritizes the interpretive role of readers, both implied and actual, in constructing literary meaning. This thesis conceptualizes characters as literary devices, with characterization as the active process of constructing these devices. While this research does not empirically investigate actual readers, it underscores the significance of reader-text interactions in characterization. Drawing on reader-response theory, this thesis argues that character sentences function as textual triggers that prompt readers' interpretive activities, enabling readers to construct nuanced understandings of characters and thus actively participate in and complete the process of literary characterization.

### 2.1.4 Summary

In summary, literary criticism rarely clearly distinguishes characters from characterization, reflecting diverse conceptualizations of characters as textual constructs, narrative functions, or simulated persons. Margolin (1983), drawing from Aristotle's foundational emphasis on action, synthesizes these perspectives by defining characters as human-like textual agents whose identities and attributes emerge primarily through their actions within narrative contexts. This action-oriented view underpins the characterization framework of this thesis. Integrating rhetorical narratology (Booth, 1983), cognitive narratology (Herman, 2009), and reader-response theory, this research positions characterization as a dynamic, triadic interaction among author, text, and implied reader. While authors deliberately embed character traits within narrative cues, implied readers actively interpret these cues to construct coherent mental representations of characters. Notably, this thesis treats reader engagement not through empirical studies of actual readers but via the theoretical construct of the implied reader, whose interpretive activities complete the characterization process. This theoretical synthesis underlies the sentence-level characterization methodology elaborated in the following chapters.

## 2.2 Computational Literary Studies on Characters and Characterization

### 2.2.1 Computational Literary Studies (CLS)

Computational literary studies (CLS) use computational methods to address questions traditionally explored through literary criticism. CLS are generally recognized as a subfield of digital humanities (DH), though some view DH itself as evolving from corpus linguistic analyses of humanities texts beginning in the mid-twentieth century (Sula and Hill, 2019). Early CLS research focused primarily on stylometry, deciphering linguistic patterns of authorship (Beausang, 2020). Later, in the 2000s, CLS broadened significantly to encompass areas such as emotion and sentiment analysis, computational narratology, fictionality, temporality, corpus building, and notably, the study of literary characters and characterization (Hamilton and Piper, 2023; Kim et al., 2020; Kukkonen and Nielsen, 2018; Mani, 2014; Milli and Bamman, 2016; Ohman et al., 2024; Piper, 2018).

Given that literary scholars long studied literature without mathematical or statistical tools, one might ask: Why adopt computational methods? Franco Moretti's landmark essay on "distant reading" (2000) significantly shaped modern DH by proposing a shift away from the traditional close reading method toward analyzing larger corpora, including vast "uncanonical" texts overlooked by literary history. Distant reading redirects attention from textual minutiae to broader analytical units such as devices, themes, tropes, genres, and narrative systems. Piper (2016), in his manifesto on cultural analytics, further identifies computational methods as essential for addressing the "generalization problem": literary criticism traditionally derives broad conclusions from analyzing a limited selection of canonical texts, inevitably overlooking the vast majority of literary production. While human readers are inherently limited by time and cognitive resources, computers simulate reading at a much larger scale, uncovering patterns in narrative arcs, writing styles, and other textual features. Piper also emphasizes computational methods for their *explicitness*, contrasting the opacity of human cognition with the transparent logical architecture of computational algorithms, even though algorithms may sometimes be criticized as "black boxes." Thus, the inclusiveness and transparency afforded by computational methods substantially enhance traditional literary scholarship.

The pipeline developed in this research directly addresses the necessity of computational method in character studies. First, by extracting *character sentences* from diverse narrative texts, the approach embraces the "great unread" (Moretti, 2000, p. 55), including world literature, genre fiction, and born-digital texts typically neglected by canon-focused studies. Second, the pipeline aligns with established cognitive models of reading: readers initially identify character-related sentences, determine the informational content of those sentences, and subsequently infer stable character traits or states (Zwaan et al., 1995; Kintsch, 1988). By structuring large language model (LLM) prompts around these cognitive stages—(i) identifying character-relevant sentences, (ii) classifying sentence types, and (iii) attributing characters—the pipeline explicitly externalizes the cognitive reading process, providing a theoretically grounded foundation for subsequent character analyses.

### 2.2.2 Core Research Tasks

Few computational literary studies directly address literary characterization (Balossi, 2014; Koukpossi and Guezohouezon, 2023). Most research instead centers on "character analysis," addressing specific character-related tasks (Jacobs, 2019; Kwon and Shim, 2017; Labatut and Bost, 2020). This section summarizes four core computational character analysis tasks: character identification, character representation, affect analysis, and character profiling.

#### 2.2.2.1 Foundational Character Identification

Character identification involves locating and identifying characters within literary texts (Barros et al., 2019). By identifying characters, researchers can extract "character-texts," the textual segments linked directly to specific characters, enabling deeper analyses (Piper, 2018).

Basic character identification often involves searching texts for explicit character names (Jacobs, 2019). However, this method overlooks aliases, pronouns, and implicit references, and depends heavily on manually provided character names, limiting its effectiveness.

To address these limitations, contemporary studies predominantly use Named Entity Recognition (NER) combined with Coreference Resolution (COREF) (Bamman, 2020; Hoque et al., 2023; Piper, 2024). NER identifies named entities (people, locations, etc.), while COREF links different mentions of the same entity. Bamman's BookNLP pipeline employs NER and COREF for character identification with a BERT-based encoder, clustering mentions into distinct characters. Yoder et al. (2021) improved this approach with the FanfictionNLP pipeline with spanBERT—an improved BERT-based encoder, achieving a character coreference F1 score of 71.4. Similarly, Jahan and Finlayson (2019) proposed a narratology-informed pipeline with coreference, semantic role labeling, and NER using support vector machine (SVM), achieving F1 scores of 0.81 and 0.99 on two distinct datasets.

Event detection (ED), closely related to character identification, typically categorizes textual events into activities, achievements, and states (Sims et al., 2019; Vendler, 1957). Event detection is also used to detect and analyze character actions since action/activities are counted as literary events. Piper (2024) specifically identifies character actions, illustrating their distribution and embodiment throughout narratives with BookNLP's event detection module.

Quotation attribution, identifying and attributing speech to characters, forms another critical task. Pipelines like BookNLP and FanfictionNLP integrate syntactic identification of quotations and COREF-based attribution (Vishnubhotla et al., 2023; Yoder et al., 2021). Recent approaches enhance this by incorporating character embeddings (Michel et al., 2024) and large language models (LLMs), including LLaMA3, GPT3.5-turbo, GPT4 (Chen et al., 2023; Michel et al., 2025; Su et al., 2024), achieving improved accuracy. Quotation attribution results support studies in character networks and emotional dynamics (Kubis, 2021; Kwon and Shim, 2017; Vishnubhotla et al., 2024).

Although these fundamental character identification tasks cannot directly address characterization, they pave the way for downstream analysis, which could go further beyond statistics of character presence, such as character embedding, social network, and emotion modeling. In this research of CS, the identification of character presence, events (actions), and dialogues is also an important step, however, they have been built in an end-to-end workflow of CS extraction.

#### 2.2.2.2 Character Representation

Character representation transforms narrative figures into structured, machine-readable formats that capture their most salient traits and relationships. Typically, a character is encoded either as a numeric vector (embedding) that establishes a compact profile of descriptive features or as a relational graph (network) that maps connections with other characters. These representations enable algorithms to compare and cluster characters or to feed character information into downstream models, making large-scale, data-driven analysis of literature possible.

Character embeddings extend word embeddings by combining all textual references of a character into a single high-dimensional vector representing their narrative portrayal. Bamman et al. (2014) pioneered this approach by combining word embeddings with hierarchical Bayesian modeling. Inoue et al. (2022) introduced innovative graph neural network-based embeddings and low-dimensional embeddings derived from character occurrence patterns. Other recent advancements include Yao’s (2023) neural network-based embeddings achieving high accuracy in predicting characters’ gender. Yang and Anderson (2024) designed AustenAlike, a benchmark for evaluating character similarities by calculating the cosine similarity with character embeddings.

Character networks symbolically represent relationships between characters through their interactions. Many studies construct networks from dialogue or co-occurrence within short textual windows (Algee-Hewitt, 2017; Elson et al., 2010; Jayakumar et al., 2022; Kubis, 2021). Jayannavar et al. (2015) explore interaction and observation networks, while Sims and Bamman (2020) investigate information propagation within literary character networks.

#### 2.2.2.3 Affect Analysis

Affect analysis broadly encompasses computational studies of sentiment and emotion within texts (Nandwani and Verma, 2021). Sentiment analysis assigns polarity scores (positive, negative, neutral), whereas emotion detection classifies text into discrete emotion categories like happiness, anger, or sadness. Complex affect analyses include emotion dynamics and character-emotion relationships (Kim and Klinger, 2018a).

Two psychological models guide affect analysis: dimensional (e.g., valence-arousal) and categorical (e.g., Ekman's six basic emotions). Dimensional approaches, such as Russell's circumplex model, measure emotion along continuous axes. Nalisnick and Baird (2013) and Jacobs (2019) employ dimensional models to map sentiment and emotional complexity in literary characters. Categorical models, as used by Moreno-Jiménez et al. (2020), classify emotion into discrete categories like anger, fear, joy, sadness, and disgust, often extending analyses to identify emotion triggers and experiencers (Jhavar and Mirza, 2018; Kim and Klinger, 2018b, 2019). The emotional characteristics of the character are represented by plotting their emotions on a two-dimensional valence-arousal axis or a categorical radar graph.

#### 2.2.2.4 Character Profiling

Character profiling synthesizes detailed portrayals of characters, summarizing traits and attributes from diverse sources and analytical signals. Situated between identification and representation, profiling helps readers achieve a holistic understanding of characters, enabling richer comparative analyses.

Personality profiling, drawing on psychological frameworks such as the Big Five (OCEAN) model, is a prevalent research area. Studies by Berov (2017), Egloff et al. (2018), Flekova and Gurevych (2015), and Jacobs (2019) apply linguistic features to profile character personalities. Rusnachenko and Liang (2025) adopt spectrum-based profiling, while Jayakumar et al. (2022) model temporal variations in characters' emotions and social behaviors.

Emotion profiling emphasizes comprehensive presentations of character emotions across narratives. Jacobs (2019) uses emotion spaces, and Jhavar and Mirza (2018) developed EMOFILE, a system mapping character emotions through narrative progression.

Profiling can also focus on specific character features. Hoque et al. (2023) developed PORTRAYAL, visualizing character attributes such as sentiment and actions. Bamman et al. (2013, 2014) explore character personas via latent topic models. Piper (2024) categorizes character actions into cognitive and embodied dimensions, while Jaipersaud et al. (2024) employ LLMs to examine implicit character attributes. Profiling further serves as a metric for evaluating narrative understanding in LLMs (Yuan et al., 2024).

#### 2.2.2.5 Summary

This section reviews key CSL investigations into literary characters, spanning foundational tasks such as character identification and representation as well as downstream applications like emotion analysis and personality profiling. Despite impressive progress in character understanding, prior work often treats the text itself merely as a corpus, an undifferentiated input, or an annotation substrate, rather than as evidence warranting close scrutiny. To address this gap, the present study focuses on isolating and extracting the specific narrative spans that portray characters, providing a firmer textual basis for subsequent analyses of characterization.

### 2.2.3 Granularity and Methods

Granularity, originally a concept from physics measuring particle size, in computer science and natural language processing (NLP) refers to the scale or hierarchical level at which data or textual information is analyzed and processed (Wang et al., 2017). In computational literary studies, granularity specifically denotes the textual segmentation level, ranging from tokens and sentences to chapters, entire passages, or complete works, at which analytical methods are applied. The choice of granularity significantly influences the narrative phenomena that can be detected. For instance, token-level analysis typically employs methods such as word embeddings and cosine similarity to measure lexical patterns, whereas chapter- or book-level analyses might aggregate sentiment scores or utilize pretrained language models to summarize character arcs or themes. Theoretical arguments from narratology also underscore the importance of granularity; Ohmann (1966), for example, posits that the sentence often serves as the basic meaning-bearing unit of literary texts. Similarly, computational studies have demonstrated that selecting an appropriate analytical scale is crucial, as being “too close(ly) to events” may hinder accurate depiction of character relationships (Elsner, 2012, p. 634). Thus, clarifying the granularity at which computational analyses occur helps ensure alignment between analytical methods, narrative theories, and literary interpretation.

This survey categorizes current computational character studies into three granularity levels: token-level, sentence-level, and passage/book-level.

In the context of NLP, a token is a fundamental unit that algorithms can process. In English, normally, short words are represented as a single token, while longer words may be split into two or more tokens (i.e., "darkness" may be split into two tokens of "dark" and "ness"). Token-level analysis in computational literary studies extends beyond mere word frequency counts, underpinning more nuanced character interpretations. Annotation tools such as BookNLP and FanfictionNLP provide token-level annotations such as named entities and coreference resolution, enabling sophisticated comparative analyses, including comparisons between canon and fanfiction characters (Milli and Bamman, 2016). Piper (2024) leverages BookNLP's token-level output to analyze embodied agency in literary characters. Jacobs (2019) computes characters' emotional complexity and personality traits by measuring cosine similarities between character-name embeddings and predefined emotion or personality word lists. Inheriting from corpus-linguistic methods, Archer and Gillings (2020) identify statistical keywords using log-likelihood and log-ratio metrics, alongside part-of-speech (POS) tagging, to examine Shakespeare's portrayal of deceptive characters.

Sentence-level analysis exists in computational literary studies for the study of sentiment, emotion, or multilingualism; however, it is rare in the context of character studies. Some studies focus on character-related texts, yet their analyses are not strictly at the sentence level. Piper (2018) defines "character-text" as referring to all words on a dependency tree linked to a character entity token; although the dependency trees are created based on sentence structures, the analytical unit remains a token. Quoted dialogues and play lines are usually treated as one kernel, although they may contain multiple sentences (Kwon and Shim, 2017; Michel et al., 2025; Vishnubhotla et al., 2023). Given the volume of information this kernel contains, quoted dialogues are closer to sentence-level analysis than token-level, passage-level, or book-level analysis. Concerning the special form of quotes and dialogues, this research treats a full quotation enclosed with quotation marks as one character sentence.

At broader narrative scales, researchers study characters by aggregating information across extended textual spans—passages, chapters, or entire books. Character representation studies, including character embeddings and network analyses, often operate at this scale. For example, Jayakumar et al. (2022) introduce role-sequence embeddings that encode each character's actions throughout the plot. Yang (2022) employs graph neural networks to propagate character traits across multi-book series, demonstrating the utility of book-level aggregation for understanding character development. Other studies focus on emotional trajectories; Reagan et al. (2016) analyze emotional arcs by applying sentiment analysis to full-length English novels, identifying six basic emotional arc shapes that dominate storytelling.

Context is essential to narrative understanding. While token-level analyses provide keywords for characters, the outputs often require substantial manual interpretation and may introduce analytical bias. Token-based analysis also risks fragmenting coherent meanings. Conversely, passage- and book-scale processing offers a holistic understanding of characters but heavily depends on the contextual window permitted by the corresponding method. Sentence-level analysis, however, offers a balanced approach, providing sufficient contextual information about tokens without aggregating too many characters, actions, or scenes. Therefore, this thesis adopts sentence-level analysis as the analytical scale for characterization, aiming to capture nuanced and contextually rich portrayals of characters.

### 2.2.4 Datasets

Domain-specific datasets are essential for developing and benchmarking computational methods in literary studies. Bamman et al. (2019) highlight that named entity recognition models trained primarily on news corpora, which often have male-centric biases, achieve low accuracy (only 38%) in recognizing female character mentions in literary texts. This section surveys datasets specifically tailored for computational character analysis.

One foundational dataset is LitBank (Bamman, 2020), which serves as the training and benchmarking corpus for BookNLP. Derived from public-domain novels available through Project Gutenberg, LitBank includes annotations for named entities, coreference, quotation attributions, and narrative events. Bamman et al. (2013) also introduced a character-focused dataset derived from movie plot summaries on Wikipedia, comprising 42,306 summaries accompanied by detailed metadata about films and characters.

The REMAN dataset (Relational EMotion ANnotation) addresses emotion recognition through semantic role labeling (Kim and Klinger, 2018a; Klinger et al., 2020). REMAN samples literary texts from Project Gutenberg and annotates entries within a three-sentence window, marking emotions, emotion triggers, experiencers, targets, and causes.

An important recent contribution, PDNC (Project Dialogism Novel Corpus), is currently the largest available dataset for literary quotation attribution (Vishnubhotla et al., 2022). PDNC provides detailed annotations including speakers, listeners, quote types (explicit, anaphoric, implicit), reporting clauses, and coreference mentions. It serves as a benchmark for state-of-the-art models and LLMs tackling quotation attribution tasks (Michel et al., 2024, 2025; Su et al., 2024; Vishnubhotla et al., 2023, 2024). Sims and Bamman (2016) present a smaller quotation attribution dataset consisting of 1,765 quotations sampled from 100 literary texts. Additionally, Chen et al. (2023) offer a specialized Harry Potter dialogue dataset, labeling dialogue scenes comprehensively with character relations, dynamics, and narrative structures.

Several datasets explicitly target character understanding tasks. Brahman et al. (2021) introduce the LISCU dataset, which supports character identification and description generation by pairing literary excerpts with summaries and detailed character descriptions. The BOOKWORM dataset, combining texts from Project Gutenberg with expert-generated character analyses to benchmark the ability of LLMs to comprehend and describe characters from book-length contexts (Papoudakis et al., 2024).

Yang and Anderson (2024) developed AustenAlike, a dataset focused on character similarity assessments in Jane Austen's novels. AustenAlike benchmarks character similarity across social attributes (rank, wealth, gender), narrative roles (heroine, hero, deceiver), and expert judgments, and evaluates the character similarity performance of BookNLP, FanfictionNLP, and GPT-4-generated rankings.

These datasets illustrate the rich diversity in computational character analysis, facilitating benchmarking for tools ranging from BookNLP to contemporary LLMs. However, no current dataset has explicitly targeted sentence-level character portrayal text retrieval; for the analyses that require character-specific texts rather than a whole book, scholars still must devise ad-hoc heuristics or manual filters to locate portrayal-related passages initially. This thesis addresses this problem by introducing the HPCS dataset, specifically designed to enable sentence-level investigations into character portrayal, thus providing a critical resource for future research and methodological innovation in computational literary studies.

### 2.2.5 Large Language Models (LLMs) in Character Studies

Large language models (LLMs) are AI systems trained on extensive textual data, capable of coherent text processing and generation, and versatile across various tasks (Naveed et al., 2024). NLP methods evolved from early statistical methods to pretrained language models (PLMs) like BERT, and subsequently to LLMs, which possess significantly larger model parameters (billions) and training datasets (gigabytes to terabytes). Leading LLMs include GPT-4o, LLaMA-3, and Gemini-2.5.

Prompt engineering involves designing and refining user instructions (prompts) to guide LLM outputs effectively across diverse tasks (Meskó, 2023). Unlike fine-tuning, which demands task-specific datasets and high computational costs, prompt engineering offers simplicity, adaptability, and strong domain-specific and general task performance (Vijayan and Vengathattil, 2025). However, prompt-engineered LLMs may not outperform standard NLP benchmarks in all tasks (Hicke and Mimno, 2024; Pagel et al., 2024). Common prompting strategies include:

- **Zero-shot prompting:** supplying only the task description with no example input and output, relying on the model's pretrained knowledge.
- **Few-shot prompting:** adding a handful of exemplars of input-output pairs to illustrate the desired pattern.
- **Chain-of-thought (CoT):** cueing the model to articulate intermediate reasoning steps before the answer, often initiated with a "please think step by step" sentence.
- **Rule-based or constraint prompting:** embedding explicit instructions or allowed formats that the model must follow.

These prompting techniques are often umbrellaed under the term in-context learning (ICL), which is often interchangeable with prompt engineering (Dong et al., 2024). The rapid development of LLMs offers new possibilities for literary analysis, including fundamental NLP tasks and complex narrative understanding (Carroll, 2024; Hicke and Mimno, 2024; Zhu et al., 2025). This section reviews LLM applications specifically within literary character studies.

Researchers have applied LLMs to fundamental NLP tasks such as named entity recognition (NER) and coreference resolution (COREF). Hicke and Mimno (2024) fine-tuned LLMs (T5, mT5, Pythia) on the LitBank dataset, with T5-3b achieving the highest performance (F1 scores: NER 91.03%, COREF 80.16%). Yet, the researchers point out that BookNLP can handle dialect texts and implicit entities more effectively. Zhu et al. (2025) introduced LLM-Link, a dual-LLM system employing memorization and prompt optimization, achieving even higher results (NER 98.4%, COREF 81.5%) on LitBank.

Quotation attribution has also leveraged LLMs. Chen et al. (2023) assessed LLM capability using the Harry Potter Dialogue dataset, noting significant gaps compared to human expertise. Michel et al. (2025) utilized prompt-based quotation attribution with LLaMA-3-8b, reaching 90.6% accuracy on the $\text{PDNC}_1$ dataset. Such tasks are fundamental to character identification and downstream analysis.

LLMs support studies on character understanding and attributes. Baruah and Narayanan (2024) used Flan-T5 and GPT-3.5 for attribute extraction from movie scripts, achieving highest accuracy (78%) with few-shot prompting. Jaipersaud et al. (2024) introduced LIIPA, using LLMs to classify characters' implicit traits (intelligence, appearance, power). Jang and Jung (2024) compared Claude-3.5-Sonnet and GPT-4o with human experts on Korean sci-fi novels, demonstrating substantial labeling accuracy (over 65%) despite differing selections. Yu et al.

(2024) developed the Literary Fiction Evaluation Dataset (LFED), benchmarking six LLMs on character relationships and characterization, where ChatGPT performed best with zero-shot prompts. Yuan et al. (2024) evaluated LLM-generated character profiles, noting promising accuracy and consistency.

LLMs have enhanced accuracy for foundational annotation tasks (NER, COREF, quotation attribution), surpassing traditional methods. Improved accuracy at these tasks facilitates higher-quality downstream analysis (emotion analysis, relationship extraction). The extended context capabilities of LLMs support nuanced, long-range narrative analysis. Additionally, prompt engineering and API integrations reduce technical barriers for literary scholars, enabling broader computational literary research. Furthermore, generative capabilities allow LLMs to simulate human interpretation, fostering innovative hermeneutic explorations. Given these advantages, this thesis utilizes LLMs as the primary analytical tool.

Annotation contamination and hallucination pose significant challenges to the reliability of LLMs. Annotation contamination occurs when evaluation data overlaps with pretraining datasets, potentially biasing results (Magar and Schwartz, 2022). This was investigated through a masking task in quotation attribution and found minimal impact on the high accuracy of LLaMA-3 (Michel et al., 2025). *Hallucination* in the context of LLMs refers to the fabrication or distortion of information output by LLMs. Zhu et al. (2024) effectively mitigated hallucination by incorporating automatic prompt optimization and a dual-LLM architecture with memorization, significantly enhancing reliability. Additional concerns when integrating LLMs in literary studies include the substantial costs of accessing closed-source LLMs and the extensive GPU resources required for their fine-tuning. Moreover, benchmarking new analytical tasks demands the creation of novel datasets and benchmarks, often requiring considerable human effort.

Aside from the technical concerns, critical problems in the methodology persist. Echoing earlier reflections on granularity, the sentence serves as an analytical unit that captures both its theoretical significance in literary study and the amount of context it naturally provides. However, limited research employs sentence-level analysis, especially sentence-level information extraction. For instance, Jaipersaud et al.'s (2024) sentence-level implicit characterization model (LIIPA-sentence) relies on manually preset sentences rather than automated extraction from coherent texts. Jang and Jung (2024) found poor alignment of LLM-generated text-span boundaries with human annotations (GPT-4o: 42%, Claude-3.5-Sonnet: 64.8%), highlighting the need for advancements in accurate information retrieval and sentence segmentation. Thus, an LLM system capable of effectively retrieving and aligning characterization-related sentences does not currently exist. Addressing these limitations, this thesis introduces an LLM-based framework with predefined sentence boundary modules, aiming to enhance the automated, context-sensitive extraction of character sentences from coherent literary texts.

### 2.2.6 Summary

Computational literary studies (CLS) on characters and characterization apply quantitative methods that align with the "distant reading" approach, offering both inclusiveness across large literary corpora and transparency in analytical reasoning. Core research tasks include character identification (through named entity recognition, coreference resolution, and quotation attribution), character representation (via embeddings and networks), affect analysis, and character profiling. Analytical granularity varies across token-level, sentence-level, and passage/book-level analyses, with sentence-level approaches being less common despite their capacity to balance contextual richness and computational tractability.

This section also reviewed existing datasets used to benchmark character analysis tasks, highlighting the diversity of resources but also the absence of sentence-level datasets for character portrayal. Additionally, the survey examined the role of LLMs, recognizing their substantial contributions to enhancing task performance and lowering technical barriers for literary scholars, while also acknowledging challenges such as annotation contamination, hallucination, computational demands, and unresolved methodological problems. These problems, particularly in sentence-level extraction, motivate this thesis's design of an LLM-based pipeline that incorporates predefined sentence boundaries to automate the extraction and analysis of character sentences from full literary texts.

### 2.3 Synthesis and Positioning

Prior research has established a robust theoretical and computational foundation for studying literary characters. Building upon this body of work, the present thesis advances computational literary studies (CLS) of characterization in four key aspects:

Firstly, from a theoretical standpoint, contemporary cognitive narratology emphasizes characterization as a cognitive process wherein readers interpret textual cues to assign attributes to characters. Yet, existing research has largely overlooked systematic exploration and extraction of texts specifically oriented toward characterization. This thesis directly addresses this gap by identifying and extracting sentence-level characterization texts and attributing these character sentences to the corresponding characters, thus bridging cognitive narratology's theoretical insights with empirical computational methods.

Secondly, current CLS approaches typically concentrate either on foundational tasks, such as character identification and representation, or on downstream analyses like affective analysis and character modeling. Although certain studies necessitate the identification of character-related texts (e.g., Jang & Jung, 2024; Piper, 2018, 2024; Tayal et al., 2022), these often lack clear criteria for text selection or default to using all sentences or passages mentioning characters. Recognizing the need for precision to avoid analytical noise, this thesis introduces the first systematic method explicitly designed for accurately retrieving characterization-related texts.

Thirdly, existing CLS on characters predominantly aggregates data either at the token level or at the passage/book level, while the intermediate sentence-level analysis remains underexplored. Given that sentences are theoretically recognized as fundamental units of narrative, providing balanced contextual granularity, this thesis proposes, defines, and operationalizes the novel analytical unit, character sentence. It provides explicit criteria, a gold-standard dataset, and an automated extraction pipeline for character sentences.

Lastly, contemporary computational character studies frequently rely on existing NLP tools such as BookNLP, spaCy, or traditional machine-learning models, all predominantly based on pretrained language models (BERT) or rule-based approaches. Recently, LLMs have demonstrated potential for character-related NLP tasks, yet their application to the specific identification of characterization texts has not been explored. This thesis pioneers the use of LLMs for characterization text identification, offering a novel methodological direction to the field.

## 2.4 Conclusion

This chapter provides a literature review on the theoretical and computational foundations relevant to the study of literary characters and characterization. The narrative theories section outlined how classical narratology, psychological models, and cognitive narratology conceptualize characters and characterization, emphasizing the triadic interaction among author, text, and reader. The computational literary studies section summarized current research tasks, methods, datasets, and the emerging use of LLMs for character analysis. It also identified key research problems, particularly in the area of sentence-level characterization extraction.

Building on these foundations, this thesis proposes a theory-informed, LLM-integrated approach for extracting character sentences from full-length literary texts. Chapter 3 will introduce the conceptual and operational definition of character sentences, and chapter 4 will illustrate the detailed design of the character sentence extraction pipeline, including its dataset creation, system architecture, experimental design, and evaluation metrics.

# Chapter 3: Character Sentences: Conceptual and Operational Definition

This thesis introduces character sentences as an analytical unit for literary characterization; a definition of the concept guides the identification of character sentences for both human readers and computational algorithms.

This chapter first presents a conceptual definition (§3.1), outlining theoretical foundations and categories of character sentences. Next, an operational definition (§3.2) translates the conceptual framework into practical criteria for identification and attribution in literary texts. Finally, §3.3 revisits both definitions to clarify their interrelation and underscore how the operational criteria are grounded in conceptual theory—an essential step for ensuring methodological transparency in the following NLP pipeline design (Chapter 4).

## 3.1 Conceptual Definition

A *character sentence* in this research is defined as a sentence that offers information about or pertains to a specific literary character, enabling readers to ascribe attributes to the character. These attributes include mental states (beliefs, desires, intentions), physical traits (appearance), and social or affective impressions (emotional responses they evoke in others). The information provided by the sentence includes two categories: **actions** and **descriptions**.

### 3.1.1 Theoretical Framework

Although linguists have developed theories concerning sentence functions and categories in fictional texts (Adam, 2011; Banfield, 2014), there is no unified theory specifically elaborating sentences that function as characterization units in fiction, particularly from a literary writing perspective. Existing materials on sentence functions related to characterization are often practitioner-based summaries supported by linguistic and literary theories. The current research synthesizes and refines these summaries into a systematic categorization.

One influential conceptualization widely used in literary writing and education is the distinction between direct and indirect characterization. *Direct characterization* explicitly describes a character's physical or personality traits, whereas *indirect characterization* conveys character traits implicitly through dialogue, actions, and responses to situations (Heier, 1987). The roots of this distinction can be traced back to early 20th-century literary discourse on "showing versus telling," first articulated by Percy Lubbock (1921), and further developed by E. M. Forster (1927) and Wayne C. Booth (1983). A complementary narrative technique, "show, don't tell," commonly attributed to playwright Anton Chekhov, underscores the importance of illustrating character traits through detailed, sensory-rich descriptions rather than explicit authorial commentary (Chekhov, 1999).

Fiction writers have also contributed practical frameworks categorizing the functions of sentences. For example, science fiction author Michael La Ronn (2021) classifies sentences into five categories: characters' opinions on settings or situations, backstory, actions, dialogues, and sensory details. Another writer categorizes fictional sentences into summary, description, blocking (actions), dialogue, and introspection (characters' thoughts) (Liz, 2024). While these frameworks are not explicitly labeled as "characterization," their categories are clearly oriented toward character development rather than plot or thematic exposition.

Addressing the theoretical gaps and drawing upon both academic and practitioner-based insights, this thesis categorizes characterization sentences into two overarching groups: actions and descriptions. The two categories effectively encapsulate essential elements identified consistently across literary, linguistic, and practitioner frameworks. While direct characterization emphasizes descriptions, "action" is broadly recognized as pivotal across theories, stemming from Aristotle's argument that actions fundamentally define and generate character (Walcutt, 1966). Contemporary computational literary studies similarly emphasize the critical role of action analysis in quantitative studies of characterization (Hoque et al., 2023; Jahan and Finlayson, 2019; Piper, 2024).

### 3.1.2 Categories

Character sentences include both **actions** and **descriptions**, each with distinct subcategories:

- **Actions:** Physical movements or activities performed by characters, typically indicated by verbs.
  - **Verbal Actions:** Direct or indirect speech where a character speaks aloud or is quoted by a narrator.
  - **Mental Actions:** Internal cognitive and emotional processes such as thinking, realizing, remembering, deciding, or feeling emotions.
  - **Physical Actions:** Observable bodily movements, interactions with objects, and facial or gestural expressions.
- **Descriptions:** Factual, static statements about characters, not involving direct actions or movements.
  - **Personal Attributes:** Appearance, personality traits, and social, hierarchical, or institutional status.
  - **Character Relationships:** Family and social connections, allegiances, and rivalries.

- **Emotional Status:** Explicit, externally observed emotional and mental status described by another character or the narrator.

Table 3.1 below provides representative examples for each category.

**Table 3.1 Examples for each character sentence category**

| Category | Subcategory | Examples |
|---|---|---|
| **Action** | **Verbal Action** (Direct & Indirect Speech) | ***"Oh yes, everyone's celebrating, all right,"*** |
| | **Mental Action** (Thoughts, Decisions, Internal Feelings) | *For a second, Mr. Dursley didn't realize what he had seen.* |
| | **Physical Action** (Bodily Movement & Expressions) | *Dumbledore stepped over the low garden wall and walked to the front door.* |
| **Description** | **Personal Attributes** (Appearance, Personality, Status) | *Mrs. Dursley was thin and blonde and had nearly twice the usual amount of neck, ...* |
| | **Character Relations** (Family and social relations, Allegiances and rivalries relations) | *"I would trust Hagrid with my life."* |
| | **Emotional Status** (Externally Perceived Emotions) | *Fear flooded him.* |

### 3.1.3 Overlaps and Complexity

Literary sentences often fulfill multiple functions simultaneously, resulting in overlaps across these categories. For instance, verbal actions frequently occur alongside descriptive speech tags (e.g., *she cried, flinging her arms around him*), and emotional status statements often include physical actions (e.g., *Professor McGonagall sniffed angrily*). These overlaps highlight the multidimensional nature of sentence-level characterization rather than representing categorical inconsistencies. Therefore, the next section, the operational definition is introduced to provide flexible yet clear criteria, accommodating sentences that portray multiple dimensions of characterization simultaneously.

## 3.2 Operational Definition

Operational definitions are methodological tools that specify clear and replicable procedures for identifying and representing concepts. Originally developed in psychology to establish measurable criteria for abstract constructs, they have since been widely adopted across disciplines, including literary studies (Stevens, 1935; Zimmermann et al., 2007; Ng et al., 2022). An operational definition of "character sentences" is established to provide a structured guideline for the computational task of extracting character sentences and for further research.

The operational definition classifies sentences into two categories: *direct speech sentences* and *narrative sentences*. *Direct speech sentences* are explicitly quoted speech enclosed in quotation marks. *Narrative sentences* include all non-quoted sentences providing descriptive or action-related character information.

This simplified categorization prioritizes accurate extraction and attribution by avoiding the ambiguity introduced by the overlapped sentence classes. Additionally, distinguishing sentences syntactically by quotation marks ensures systematic and comprehensive identification since quotation detection and attribution is a well-established area in computer science (Michel et al., 2025; Vishnubhotla et al., 2022, 2023).

### 3.2.1 Identification

Character sentences appear in two distinct forms:

1. **Direct speech character sentences:** A full sentence or multiple sentences that are explicitly quoted speech, enclosed within quotation marks.
2. **Narrative character sentences:** A full sentence that provides character-related information outside of direct speech and is punctuated with standard sentence-ending marks, including a period (.), exclamation mark (!), question mark (?), or ellipses (…).

### 3.2.2 Direct Speech Sentences

Direct speech character sentences are defined by syntactic rules, requiring them to be enclosed within quotation marks ("..."). Each direct speech sentence should be identified and extracted with its quotation marks and attributed to its speaker.

#### 3.2.2.1 Attribution

Attribution refers to assigning identified direct speech character sentences to their respective characters. The fundamental attribution rule of direct speech sentences is that **any direct speech should be fully attributed to the speaking character.**

- **Single-Sentence Speech:** A single complete sentence enclosed within quotation marks, fully attributed to the speaker.
  - Example: *"Five, you mean, once she's taken off Hermione's."*
- **Multiple-Sentence Speech:** Two or more sentences enclosed continuously within quotation marks; these sentences are treated collectively as a single character sentence attributed to the speaker.

- Example: *“Well, I still say you were lucky, but not many first years could have taken on a full-grown mountain troll. You each win Gryffindor five points. Professor Dumbledore will be informed of this. You may go.”*

- **Multiple-Paragraph Speech:** Dialogue spanning multiple paragraphs without narrative interruption; each paragraph is identified separately but attributed collectively to the same speaker.
  - Example:

    *“To ensure that no underage student yields to temptation,” said Dumbledore, “I will be drawing an Age Line around the Goblet of Fire once it has been placed in the entrance hall. Nobody under the age of seventeen will be able to cross this line.*

    *“Finally, I wish to impress upon any of you wishing to compete that this tournament is not to be entered into lightly. ...”*

Two additional attribution rules apply in this research:

- **Collective Speech:** If multiple identifiable speakers deliver a quote collectively (e.g., *“They said,”* referring to named individuals), it is attributed to all included characters. However, quotes with unclear speaker groups (e.g., *“All Gryffindors”*) are excluded from character sentence annotation.
- **Speech Tags:** Speech tags (reporting clauses) attached to quotes to identify the speaker qualify separately as narrative character sentences and are excluded from direct speech character sentence attribution.

#### 3.2.2.2 Exclusions

**Indirect speech and free indirect discourse (FID) are excluded** from direct speech character sentences. *Indirect speech* restates another person's speech or thoughts without direct quotation, preserving the original meaning, for example: *"She asked how he got here." Free indirect discourse (FID)* integrates a character's perspective into third-person narration without explicit attribution, blurring distinctions between character and narrator (Reboul et al., 2016). Examples include:

- *Frankly, he was a happy man, thought John.*
- *She begged him to reconsider. How could he justify robbing his child, his only child, of so large a sum?*

Indirect speech and FID are excluded from direct speech character sentence for two reasons. First, for the practical concern of computationally identifying CS, indirect speech and FID are not enclosed with quotation marks, which requires additional computation for automatic extraction. There is also no current datasets or systems for FID detection. Restricting the direct speech category with only quoted speeches simplifies the identification and attribution rules, saving the input token counts and reasoning resources for the LLMs. Secondly, certain FID is implicit, easily confused with the narrator's voice (Reboul et al., 2016). To minimize the possibility of false-positive identification, indirect speech and FID are treated as narrative CS in this research, as discussed in cussed in §3.2.3. As indirect speech and FID may come with a clause identifying who is speaking/thinking, the sentence should be correctly identified and attributed as narrative CS, and future studies can further distinguish them from other narrative sentences.

### 3.2.3 Narrative Sentences

Narrative character sentences describe a character's **actions** or attribute static **descriptions** to a character outside of direct speech. These sentences contribute to characterization by detailing a character's behaviors, thoughts, emotions, or relationships through narration. Syntactically, they differ from direct speech character sentences as they are not enclosed in quotation marks.

#### 3.2.3.1 Categories and Examples

Narrative character sentences inherit the two categories of the conceptual definition: **action-based** and **description-based**.

*Action-based sentences depict characters* performing verbal, mental, or physical actions, typically indicated by verbs such as *said*, *thought*, *realized*, or *waved*. Multiple actions may co-occur within a single sentence. These actions include:

- **Verbal Actions:** Indirect speech and informative speech tags (reporting clause with additional information than *"xx said"*)
  - Example: *said Wood, frowning at this lighthearted behavior.*
- **Mental Actions:** Free indirect discourse (FID) or straightforward narration of a character's mental experiences.
  - Example (FID): *For a moment, he thought the roaring of the wind had woken him.*
  - Example (neutral narration): *Because Harry knew who that screaming voice belonged to now.*
- **Physical Actions:** Observable bodily movements, gestures, or interactions performed by characters.

- Example: *He turned his head sharply, spotting a humpbacked, one-eyed witch statue.*

*Description-based sentences* provide factual, static portrayals of characters' attributes, relationships, or externally perceived emotional states, emphasizing stable traits rather than dynamic actions. These descriptions include:

- **Personal Attributes:** A character's appearance, personality traits, or social status.
  - Example: *A long sheet of silvery-blonde hair fell almost to her waist.*
  - Example: *Mr. Dursley was the director of a firm called Grunnings.*
- **Character Relationships:** Social connections such as family ties, friendships, allegiances, or rivalries.
  - Example: *Mrs. Potter was Mrs. Dursley's sister, but they hadn't met for several years.*
- **Emotional Status:** Externally perceived emotional conditions explicitly stated by the narrator or observed by other characters.
  - Example: *Moody seemed totally indifferent to his less-than-warm welcome.*

Notably, this thesis's character sentence extraction pipeline focuses on identification and attribution, without further classifying narrative character sentences beyond the two main categories.

#### 3.2.3.2 Identification

Unlike direct speech, which is identified by quotation marks, narrative CS relies on grammatical structure, part-of-speech (POS) tags, and contextual interpretation. The following criteria guide systematic identification:

**1. Sentence completeness**

Narrative character sentences must be either complete sentences or informative speech tags.

**2. Explicit character reference**

A narrative character sentence may explicitly reference a character via names (*Harry Potter*), nicknames (*Scarhead*), pronouns (*he, she*), descriptors (*the professor*), body parts (*his eyes*), or possessions (*Hermione's book*).

**3. Implicit character reference**

Sentences implicitly referencing characters without explicit subjects, yet clearly attributable through context, also may qualify.

- Example: *Nobody seemed to want to touch him.*

**4. Action indicators**

Sentences containing verbs that indicate verbal (*said, mentioned*), mental (*thought, realized*), or physical actions (*ran, jumped*) may qualify as action-based narrative sentences.

**5. Descriptive "be" verbs**

Sentences using "be" verbs (*is, was, were*) to describe a character's personal attributes (*She was brave*), relationships (*Ron is his friend*), or emotional states (*He was angry*) qualify as description-based narrative sentences.

**6. Adjectival descriptions**

Sentences including adjectives that directly describe a character's personality (*brave, cunning*) or emotional state (*happy, fearful*) also qualify as descriptive narrative character sentences.

#### 3.2.3.3 Attribution

Narrative character sentences should be attributed to the character(s) they primarily describe or depict. The guiding principle for attribution is identifying **which character's action, emotion, or description forms the central focus of the sentence.** Below are rules to facilitate consistent attribution:

**1. Single-Character Sentences**

Sentences explicitly mentioning one character who performs an action or is described in detail should be attributed to that character.

- Example: *Harry raised his wand.*
  - **Attribution:** Harry

**2. Multiple-Character Sentences**

- **Equally portrayed characters:** Sentences describing simultaneous or parallel actions of multiple identifiable characters should be attributed to all involved characters. Exclude sentences where characters are not identifiable as individuals, since individual characters are the focus of the definition.
  - Example: *Harry, Ron, and Hermione were joined at their tray by a curly-haired Hufflepuff boy Harry knew by sight but had never spoken to.*
    - **Attribution:** Harry, Ron, Hermione
  - Example: *The audience burst into laughter.*
    - **Attribution:** Not a character sentence
- **One character acting upon another:** If a sentence highlights one character's action toward another, attribute it to the acting character. If the receiver's reaction is explicitly detailed, the sentence should also be attributed to the receiver.

- Example: *Lavender kissed Ron.*
    - **Attribution:** Lavender
- Example: *He (Harry) finally tracked her (Hermione) down as she emerged from a girls' bathroom on the floor below.*
    - **Attribution:** Harry, Hermione

- **Perspective vs. Action:** If a sentence grammatically originates from one character's viewpoint but primarily portrays another character's action or state, attribute the sentence to the portrayed character. Conversely, if the sentence emphasizes the observing character's perception or experience, attribute to the observer.
    - Example: *Harry saw Krum shake his head as he pulled his furs back on.*
        - **Attribution:** Krum
    - Example: *The trees were so thick he (Harry) couldn't see where Snape had gone.*
        - **Attribution:** Harry

**3. Reporting clause**

A reporting clause (speech tag) is a short clause attached to direct speech to indicate the speaker. Typically, it includes a reporting verb such as *said*, *yelled*, or *interrupted*, essential for computational quotation attribution (Elson and McKeown, 2010).

A reporting clause may appear before or after the speech, or interrupt direct quotations; occasionally, direct speech lacks speech tags altogether. A reporting clause can be categorized into *Minimal* and *Informative* based on its informativeness. A minimal reporting clause consists only of the speaker's name or pronoun and the verb *said*, without additional context (e.g., "*said Mr. Malfoy*"). An informative reporting clause includes alternative reporting verbs or additional descriptive information about physical or emotional states, for example:

*... he coughed.*

*... he whispered to Harry as he opened the trunk.*

*The smile faded slightly from Mr. Borgin's face.*

In this research, only informative speech tags are classified as narrative character sentences and attributed accordingly, while minimal speech tags are excluded from character sentence attribution.

**4. Free Indirect Discourse (FID)**

Free indirect discourse portrays characters' thoughts or internal experiences within third-person narration. Similar to direct speech, sentences employing FID should be attributed to the initiating character.

- Example: *His life has never been easier, he thought.*
  - **Attribution:** The character indicated by "he"

#### 3.2.3.4 Edge Cases and Exceptions

Literary texts often feature complex sentence structures, figurative language, and other stylistic devices (Bamman, 2020). This complexity can create edge cases or exceptions in identifying and attributing narrative character sentences. Three primary scenarios are outlined below.

**1. Ambiguous free indirect speech/discourse (FID)**

Some FID sentences include clear markers (e.g., "he thought," "she spoke to herself"), but others lack explicit indicators and can be mistaken for neutral narration. To reduce misclassification, follow the steps below when encountering a potential FID sentence:

- **Contextual Check:** If a questionable sentence is surrounded by explicit FID statements, infer it as FID by continuity.
  - Example: *"Dumbledore?" he said, dashing to the door to make sure. Fred was right. There was no mistaking that silver beard. Harry could have laughed out loud with relief. He was safe. There was simply no way that Snape would dare to try to hurt him if Dumbledore was watching.*
    - **Decision:** The sentence *"There was no mistaking that silver beard"* inherits the FID mode from surrounding sentences, thus identified as an FID sentence attributed to Harry.
- **Perspective Test:** If removing the sentence disrupts the flow of internal perception or thought, it is likely FID. Conversely, if a sentence interrupts an FID sequence by shifting focus to neutral narration, do not classify it as FID.
  - Example: *Ron and Hermione in the distance, jumping up and down, Ron cheering through a heavy nosebleed. Harry had reached the shed. He leaned against the wooden door and looked up at Hogwarts, with its windows glowing red in the setting sun. Gryffindor in the lead. He'd done it, he'd shown Snape ... And speaking of Snape ... A hooded figure came swiftly down the front steps of the castle.*
    - **Decision:** The sentence "And speaking of Snape ..." is not FID since this sentence breaks the FID flow of Harry, and shifts the perspective to the neutral narrative of describing a hooded figure's movement.

**2. Narrative sentences with partial quotations**

Some narrative sentences contain embedded quoted fragments that do not form complete direct speech. In such cases, treat the entire sentence as a single narrative character sentence:

- Example: *There was a loud and impatient "tuh" from behind them. It was Hermione, ...*
  - **Decision:** The first sentence is a narrative character sentence attributed to Hermione.
- Example: *Hagrid called "Who is it?" before he let them in, and then shut the door quickly behind them.*
  - **Decision:** This entire sentence is a narrative character sentence attributed to Hagrid.

### 3.3 Conclusion

This chapter establishes a conceptual and operational definition for character sentences, and a set of criteria for classifying character sentences of different kinds. The conceptual definition categorizes character sentences into actions (verbal, mental, physical) and descriptions (personal attributes, relationships, and emotional statuses). Recognizing overlaps among these categories highlights the multidimensional nature of literary sentences and characterization.

The operational definition translates these conceptual categories into practical, replicable criteria and procedures, distinguishing between direct speech and narrative character sentences. Direct speech character sentences are defined syntactically by quotation marks. Narrative character sentences, without universal punctuation patterns, are identified and attributed through linguistic and contextual cues, as well as contextual understandings. The operational definition also addresses complexities, such as free indirect discourse and speech tags, ensuring robust and systematic extraction and attribution.

Together, this chapter provides the groundwork for a comprehensive pipeline to extract character sentences and conduct computational analyses. It also informs the creation of annotation guidelines for a gold-standard dataset.

In the next chapter, the detailed design of the character sentence extraction pipeline will be introduced, covering methodology, the development of the gold-standard dataset, model selection and comparison, and an evaluation of pipeline performance.

# Chapter 4: Character Sentence Detection and Attribution: An NLP Pipeline

This chapter presents two Gold datasets and a natural language processing pipeline for character sentence extraction. Section 4.1 introduces HPCS, the Gold datasets for character sentence identification and attribution, annotated on the *Harry Potter* book series. Section 4.2 details *Ext-cs*, the character sentence identification and attribution pipeline, introducing its architecture. Sections 4.3 to 4.5 are three experiments of this thesis: comparative studies of the LLM selection, the quality metrics, and the prompt optimization of the pipeline. Section 4.6 concludes with a summary of the pipeline design and suggestions for future improvements.

## 4.1 *HPCS*: Gold datasets for character sentences

A Gold dataset is a curated collection of high-quality data that benchmarks an AI system's performance and evaluation (Kilgarriff, 1998). There is no existing Gold dataset available for benchmarking the NLP task of identifying character sentences. To address this gap, this thesis created **HPCS** (Harry Potter Character Sentences), two Gold datasets annotated from text samples of the *Harry Potter* book series. The datasets, *HPCS_clause-level* and *HPCS_full-sentence*, were annotated by the same annotators on identical book samples but differ in annotation granularity. Both of the datasets and all documentation are released in this thesis's public GitHub repository.

### 4.1.1 Data Source and Selection Criteria

The datasets are annotated on coherent narrative passages from the *Harry Potter* series by J.K. Rowling, with one sample drawn from each book. The series is selected due to its exemplary narrative complexity, well-defined characterization, and widespread adoption in NLP research (Chen et al., 2023; Liu et al., 2024; Tayal et al., 2022).

Although *Harry Potter*'s global popularity raises concerns about possible inclusion in large language models' (LLMs) training data, the tasks of character sentence identification and attribution differ substantially from standard language modeling objectives. Recent research indicates that annotation contamination does not significantly enhance model performance on similar tasks (Michel et al., 2025).

The sampling process for the datasets involved three main stages. First, the seven Harry Potter books were tokenized using the BookNLP pipeline (Milli & Bamman, 2016), which assigned paragraph, sentence, and token IDs to the text. Second, a Python script was used to randomly select five passages, each containing 200 sentences (approx. 4,000 tokens), from every book. To ensure coverage of the full narrative arc, the sentence IDs were divided into three pools representing the beginning, middle, and end of each book, and the starting sentence IDs for the passages were randomly drawn from these pools.

Finally, all sampled passages underwent manual review to ensure that each met the following criteria:

- An approximate 1:1 ratio of narrative to dialogue sentences.
- Inclusion of multiple primary and secondary characters.
- A variety of sentence structures.
- Coverage of multiple distinct scenes.

Passages that did not satisfy these criteria were replaced through resampling. The final selections were saved as *<HP[book_id]Sample.txt>*. During sampling, only the passages from

Book 5 were resampled, as all initial samples failed to meet the second criterion of character variation.

### 4.1.2 Annotation Scope

Annotations were conducted at two granularities: **clause-level** (*HPCS_clause-level*) and **full sentence-level** (*HPCS_full-sentence*).

- **Clause-level annotation:** Independent clauses separated by punctuation were individually annotated, while clauses without punctuation separation remained unsegmented.
  - Example: *Hermione muttered as Ron slipped his wand up his sleeves.*
    - **Decision:** Annotate as one unit.
  - Example: *said Malfoy loudly a few minutes later, as Snape awarded Hufflepuff another penalty.*
    - **Decision:** Annotate as two separate clauses (*"said Malfoy loudly a few minutes later,"* and *"as Snape awarded Hufflepuff another penalty."*).
- **Full sentence-level annotation:** Character sentences are annotated at sentence boundaries (periods, exclamation marks, etc.).

In both datasets, quoted dialogues were annotated as single entries, regardless of internal punctuation. Reporting clauses, even if grammatically incomplete, were also included as entries.

### 4.1.3 Annotation Guideline

A detailed annotation guideline is created based on the definitions of character sentences and iteratively refined through pilot annotations. The guideline consists of rules, examples, and notes from the annotators. The full annotation guideline is published in this thesis's GitHub repository for future usage. Here, key guidelines are presented, which include:

- **Character sentence definition:** Sentences explicitly or implicitly representing characters' verbal, physical, or mental actions, or describing their attributes, relationships, or emotions.
- **Sentence types:** Direct speech or narrative sentences (including reporting clauses and free indirect speech).
- **Identification rules:**
  - All direct speech sentences qualify.
  - Narrative sentences qualify if containing character-driven actions or descriptions.
- **Attribution rules:**
  - **Direct speech and reporting clauses:** Attribute to the speaker. If the reporting clause contains another character's action or description, attribute it to both the speaker and the acting/described character.
  - **Actor vs. receiver:** Attribute only to the grammatical subject performing an action or reaction.
  - **Multiple actors:** Attribute to all identifiable characters involved.
  - **Free Indirect Speech (FID):** Attribute to the character whose perspective or thoughts are being presented; if unclear, treat as a typical narrative sentence.
  - **Clause-level annotation only:** Segment compound sentences into independent clauses if each clause distinctly attributes to different characters.

### 4.1.4 Annotation Process

Two annotators collaboratively created the datasets. The primary annotator (the thesis author) developed guidelines and conducted training. The second annotator was hired from the Harry Potter tag (#harry potter) on the Tumblr website[1,2], which is a global fan community of the series. The secondary annotator was trained for 1.5 hours, covering guidelines, platform (Label Studio) use, and edge case handling.

Annotation proceeded in two rounds:

- **First round:** Initial annotations with uncertainty documented.
- Post-round discussion to resolve edge cases and refine guidelines.
- **Second round:** Revisions based on updated guidelines; final annotations used for calculating inter-annotator agreement.

In each annotation round, the primary annotator completed the annotations first, followed by the secondary annotator. This procedure ensured that the primary annotator was fully familiar with the annotation process and could address any questions raised by the secondary annotator.

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[1] Tumblr (https://www.tumblr.com/) is a microblogging and social media platform. The website is well-known for its hybrid tagging system; users add tags to their posts, which then form communities revolving around the tags. The tag #harry potter (https://www.tumblr.com/tagged/harry%20potter) has 2.28 million fans and over 200 daily new posts.

[2] The primary annotator published an annotator hiring post in the #harry potter tag on Tumblr, and the secondary annotator was selected from the responders. Both the primary and secondary annotators were very familiar with the book series and its characters.

Label Studio 1.15.0 (Tkachenko et al., 2020) facilitated annotation, using a text classification template with character names as categories (see Figure 4.1).

Inter-annotator agreement was calculated using Krippendorff's α (Hayes and Krippendorff, 2007), implemented via an R package *KRIPPENDORFFSALPHA* (Hughes, 2021). The results were 0.81 for *HPCS_clause-level* and 0.86 for *HPCS_full-sentence*, indicating high reliability.

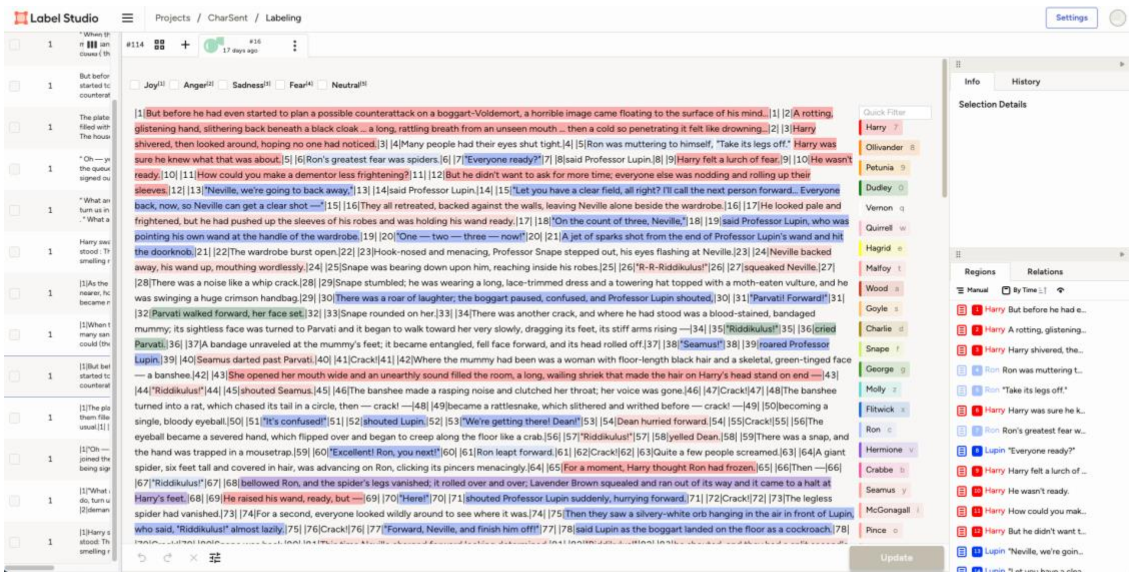


**Figure 4.1 Annotation interface of Label Studio**

### 4.1.5 Data Structure and Fields

Entries in both datasets share the following structured fields:

- ***book_id* (int, 1 to 7):** Identifier for the corresponding *Harry Potter* book.
- ***text* (str):** Annotated sentence or clause.
- ***labels* (list):** Characters attributed to the given *text*.
- ***category* (str):** Sentence type, designated either as *narrative* or *direct_speech*.

### 4.1.6 Dataset Statistics

The basic statistics of the two Gold datasets are in Table 4.1:

**Table 4.1 Statistics of the Gold datasets**

| | **Clause-level Gold data** | **Full sentence Gold data** |
|---|---|---|
| **Number of total entries** | 1459 | 1432 |
| **Number of entries per book** | Book 1: 214; Book 2: 173; Book 3: 203; Book 4: 205; Book 5: 204; Book 6: 231; Book 7: 229; | Book 1: 204; Book 2: 169; Book 3: 204; Book 4: 206; Book 5: 204; Book 6: 223; Book 7: 222; |
| **Narrative/speech ratio** | Narrative: 0.59; Direct speech: 0.41; | Narrative: 0.58; Direct speech: 0.42; |
| **Number of unique characters** | 41 | 41 |
| **Top 5 characters (with sent num)** | Harry: 514; Hermione: 264; Ron: 207; Hagrid: 57; Cho: 54[3] | Harry: 521; Hermione: 265; Ron: 261; Cho: 60; Hagrid: 57 |
| **Examples** | {"book_id": 1,"text": "As Gryffindors came spilling onto the field, he saw Snape land nearby, white - faced and tight-lipped-","labels": ["Snape"],"category": "narrative"},{"book_id": 1,"text": "then Harry felt a hand on his shoulder and looked up into Dumbledore's smiling face.","labels": ["Harry"],"category": "narrative"} | {"book_id": 1,"text": "As Gryffindors came spilling onto the field, he saw Snape land nearby, white-faced and tight-lipped — then Harry felt a hand on his shoulder and looked up into Dumbledore's smiling face.","labels": ["Harry","Snape"],"category": "narrative"} |

[3] Hagrid and Cho are both secondary characters in the Harry Potter series; however, unlike Hagrid, whose appearances are evenly distributed across the series, Cho appears primarily in Book 5. Her high frequency in the Gold dataset is due to the sample from Book 5 containing an entire scene of her first date with Harry.

A bar chart is created to demonstrate the distribution of sentence length in word counts within the Gold dataset, see Figure 4.2.

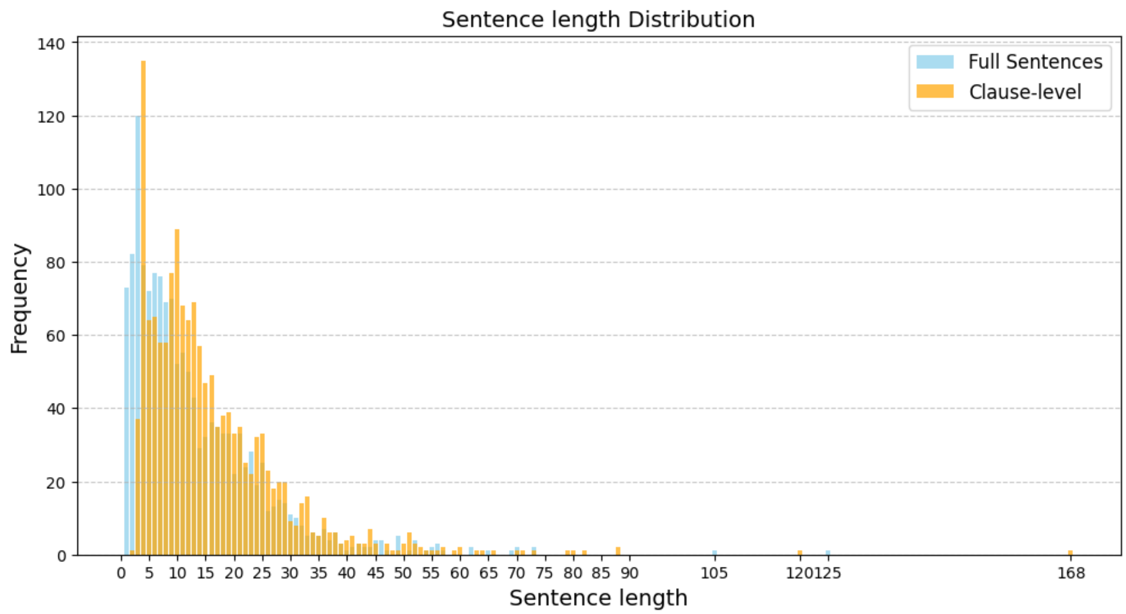


**Figure 4.2 Sentence length distribution (number of words) in the two Gold datasets**

## 4.2 *Ext-cs*: Pipeline Methodology

### 4.2.1 Overview

The purpose of the pipeline is to enable the systematic identification of character sentences in literary texts and to attribute the sentences to the portrayed characters, with the utility of large language models. The objectives of the pipeline are:

- Preprocess input literary text into clean, structured text labeled with sentence identifiers. **(Cleaning module & Senticizer module)**
- Identify character sentences using LLMs based on both content and structural features. **(LLM Processing module)**

- Attribute character sentences to the correct character(s) through structured reasoning and prompt design. **(LLM processing module)**
- Filter misattributions using syntactic parsing to increase attribution precision. **(Dependency Filter module)**
- Output structured data of character sentences for downstream analysis.

LLM integration is a highlight of the current pipeline. Existing tools for literary text analysis, such as BookNLP (Milli and Bamman, 2016) and FanfictionNLP (Yoder et al., 2021), depend on BERT-style pretrained language models, which are deep neural networks trained in advance on large corpora to predict masked words in context. These models process text in fixed-length segments (typically up to 512 tokens) and are designed for span-level predictions, which makes them effective for surface-level tasks such as named entity recognition (NER) and coreference resolution. However, these constraints limit their ability to capture broader narrative context or perform complex reasoning across longer passages. They cannot, by design, determine whether a sentence functions as characterization without additional supervised learning—a decision that demands wider narrative context and pragmatic reasoning. To overcome these constraints, this thesis inserts an LLM layer into the pipeline. The LLM (Gemini 2.5 Pro via API) can process several thousand tokens at once, reason across sentences, and directly judge (i) whether a sentence conveys characterization and (ii) which character it portrays. This single, context-aware inference step replaces a chain of brittle heuristics, enables rapid domain adaptation through prompt engineering rather than retraining.

### 4.2.2 Pipeline Architecture

The pipeline consists of four major modules, corresponding to the following five main objectives of design: the Text cleaning module, Senticizer module, LLM processing module, and Dependency filter module. See Figure 4.3 for the pipeline architecture.

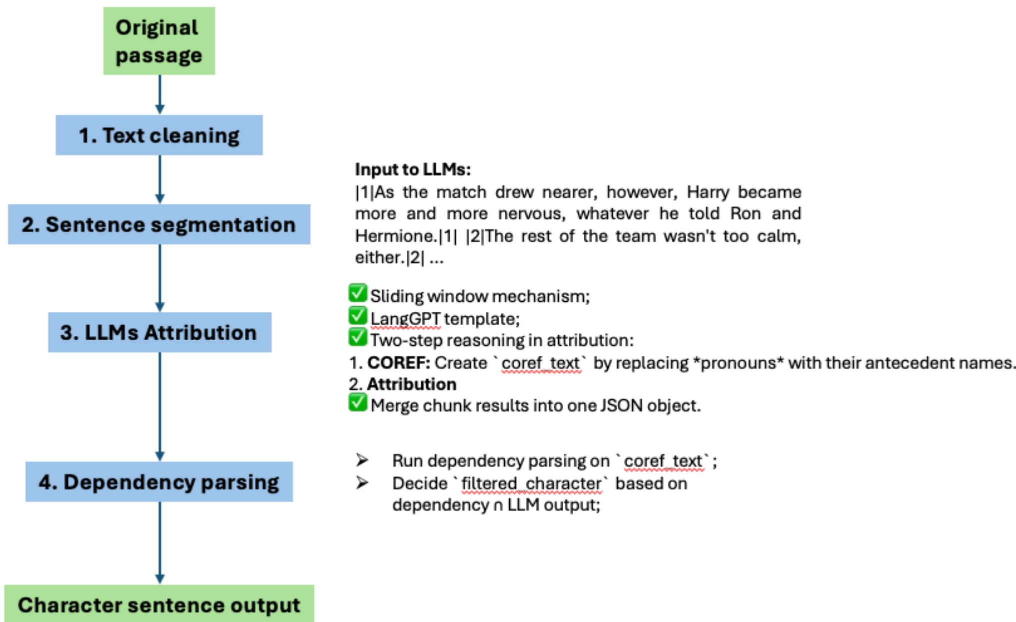


**Figure 4.3 Pipeline architecture**

#### 4.2.2.1 Input Data

The input data of the pipeline is coherent literary texts. The input data should be saved in plain text files (*.txt*). The length of the input text is unlimited, given that a sliding window mechanism will be integrated into the LLM processing module, see the section below.

#### 4.2.2.2 Text Cleaning Module (*Cleaning.py*)

The module preprocesses input texts by normalizing whitespace and managing special characters. The primary goal is to maintain the literary coherence and stylistic integrity of the text. Specifically, the module removes redundant spaces and ensures only a single empty line separates each paragraph.

**Primary functions:**

- *standarize_paragraph_spacing(text:list) -> str*: Limits paragraph separation to a single empty line.
- *normalize_whitespace(text: str) -> str*: Removes extra spaces and ensures consistent spacing.
- *handle_special_characters(text: str) -> str*: Cleans or normalizes special characters (i.e., OCR errors) without affecting narrative coherence.

#### 4.2.2.3 Senticizer Module (*Senticizer.py*)

Accurate sentence boundary identification avoids LLM from over-segmenting the input text, maintaining the pipeline output at a full-sentence level. This module segments the input text into individual sentences that aligns with the definition of character sentences, and inserts sentence ID markers at both the beginning and end of each sentence to clearly delineate sentence boundaries (Michel et al., 2025). The module leverages Python's *spaCy* with customized rules and regular expressions for sentence segmentation. The customized rules are consistent with the definition of character sentence and the Gold data annotation scheme:

- **Dialogues within quotation marks**: Considered as complete sentences.
    - If dialogue spans multiple paragraphs without interruption (each non-final paragraph begins with a quotation mark but does not end with one, e.g., *"xxx.*), treat each uninterrupted quoted paragraph as a single sentence.
    - Ignore quotation marks used for cited interjections and segment these sentences based on regular sentence-ending punctuation.

- **Reporting clauses connected to dialogue**: Identify and label reporting clauses as independent sentences.
- **All other sentences**: Segmented according to standard punctuation marks (*.*, *!*, *?*).
  - Ellipses (*...*) and colons (*:*) are also treated as potential sentence boundaries if the clause next to the ellipses or colons begins with a capitalized word.

**Primary functions:**

- *split_non_quote_sentences(text: str) -> list*: Adds rules to *spaCy*'s *senticizer*, split non-quote sentences by*:* and *...*.
- *quotation_aware_split(text: str) -> list*: Handles the quotation rules.
- *process_and_save(text: list) -> file*: Saves the segmented text into a file with sentence ID markers.

**Example output:**

*|210|"So what on earth's Hagrid up to?"|210| |211|said Hermione.|211|*

*|212|When they knocked on the door of the gamekeeper's hut an hour later, they were surprised to see that all the curtains were closed.|212| |213|Hagrid called "Who is it?" before he let them in, and then shut the door quickly behind them.|213|*

#### 4.2.2.4 LLM Processing Module (*RunGemini.py* & *MergeChunkResults.py*)

The LLM processing module employs state-of-the-art LLMs to identify and attribute character sentences within coherent literary passages. In the current pipeline, **Gemini 2.5 Pro** (*gemini-2.5-pro-exp-03-25*) is selected as the working LLM due to its high performance, fast speed, and low cost (Bercovich et al., 2025).

**1. Sliding window mechanism:**

A sliding window mechanism is implemented to restrain the LLM input within the token limitation of Gemini 2.5 Pro (input: 1,048,576 tokens; output: 65,536 tokens). The sliding window segments the input text into overlapping chunks (*window_size = 1500*, *overlap = 300*). If character sentence attributions in overlapping segments conflict, only the output from the first segment is retained.

**2. Prompt design:**

The prompt follows the **LangGPT prompt framework**, enhanced with **two-stage coreference reasoning**. LangGPT is a dual-layer prompt design structure shown to significantly improve LLM performance (Wang et al., 2024). The prompt includes:

- *<Role>*: Assigns the LLM the role of a literature expert.
- *<Profile>*: Specialization in sentence-level characterization and character analysis.
- *<Goals>*: Identify character-related sentences and attribute them to the correct characters.
- *<Rules>*: Includes quotation attribution, reporting clause handling, multi-character attribution, etc.
- *<Workflow>*: Guides step-by-step reasoning.
- *<Initialization>*: Integrates prompt sections into one directive:

*As a <Role>, your goal is to <Goals>. You must follow the <Rules> and <Workflow>. No extra output beyond the final JSON.*

A **two-stage Coreference Reasoning** (included in *<Rules – Narrative Sentences>*) is added to reduce coreference ambiguity and enhance the model's accuracy in attribution:

- **Stage 1 – Coreference Expansion**: Replaces ambiguous pronouns in each sentence with their antecedents to form a *coref_text* string.

- **Stage 2 – Who-Acts Filter**: Ensures only the grammatical subject (i.e., the actor) is included in the *character* list, based on the definition of character sentences. This avoids attributing actions to recipients or passive entities.

For the full prompt, see Appendix A.

A **character dictionary** is preset to avoid LLM hallucination in the attribution task. The keys in the character dictionary are character names, and the value is a dictionary that contains the variants of the character name. See an example below. The character dictionary is created manually to restrict the LLM's output to the characters that the users want to focus on.


```
{
    "Harry":{
            "names":[
                    "Harry",
                    "Harry Potter",
                    "Mr. Potter",
                    "Mr. H. Potter"]
    }
}
```


Notably, the current module only performs identification and attribution on **full sentences**, as defined by the preceding segmentation module. Clause-level segmentation is explicitly excluded at this stage to preserve structural consistency.

**3. Concatenate LLM output:**

Structured output is enabled via the Gemini API, returning JSON objects with the following fields:

- *text*: Original sentence
- *coref_text*: Sentence with expanded coreference

- *type*: Sentence type (*narrative* or *direct_speech*)
- *character*: List of attributed characters

A Python script (*MergeChunkResults.py*) is integrated into the module to concatenate all chunk results of the sliding window and return a JSON object, removing the duplicated entries. Invalid character names in the *character* field are handled by looking up the character dictionary; if an entry's *character* is not in the keys or values of the character dictionary, remove the entry. Duplicated and invalid character entries will be reported to the user for quality control purposes; if there are too many reported entries, the LLM might have hallucinated, and the user should try re-running the pipeline on the specific input.

**Primary Functions**

- *sliding_window_chunks(text: str) -> list[str]*: Segments input text into overlapping chunks using a sliding window, ensuring each chunk fits within the LLM's input and output token limits.
- *load_prompt(file) -> str*: Loads the full prompt template and character dictionary, which constrains the character names allowed in the LLM output.
- *_generate_once(text: str) -> str*: Sends a single prompt request to the LLM and returns the raw structured output.
- *parse_and_save(text: str) -> file[json]*: Parses the structured output from the LLM into a JSON object and saves it to a file.

**Example Output**


```
[
    {
            "book_id": 2,
            "text": "Her words were cut short, however, as the portrait of the fat lady swung open
and there was a sudden storm of clapping.",
```

"coref_text": "Hermione's words were cut short, however, as the portrait of the fat lady swung open and there was a sudden storm of clapping.",
"type": "narrative",
"character": [
"Hermione"
]
},
...
]

#### 4.2.2.5 Dependency Filter Module (*Dependency.py*)

Experimental results indicate that even with the "who-acts filter" implemented in the prompt, LLMs often include both the doer and the receiver of an action in the *character* list. To address this, a dependency parsing filter module is added after the LLM processing step to refine the *character* list based on grammatical roles.

Dependency parsing is an NLP technique that analyzes the grammatical structure of a sentence by identifying the relationships between words. By applying dependency parsing, each word and punctuation within the sentence will be assigned a dependency tag to indicate their grammatical position. The analysis is as illustrated in Figure 4.4.

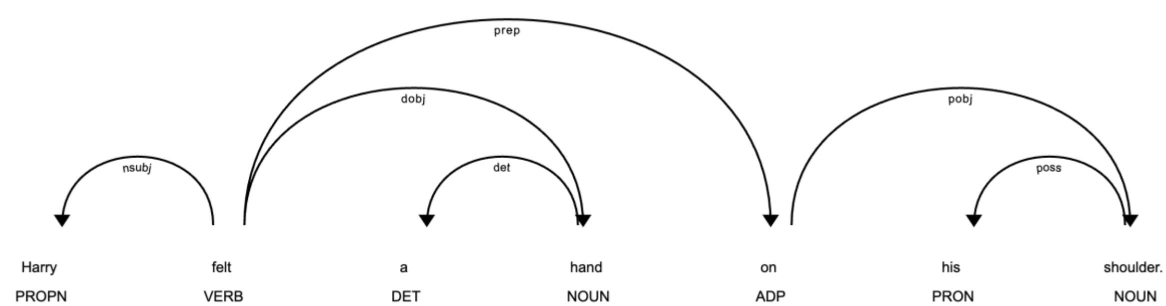


**Figure 4.4 Dependency parsing**

This module uses spaCy's *en_core_web_trf* model to extract dependency tags for each sentence. Parsing is performed on the *coref_text* field from the LLM output. Only character mentions with the dependency tags *nsubj* (nominal subject) or *nsubjpass* (passive nominal subject) are retained. However, no *nubjpass* tag was encountered in the pipeline design and assessment process, which means that there were no passive sentences in the current Gold dataset. The intersection of the LLM-attributed *character* list and the dependency-identified subjects is stored in a new field called *filtered_character* in the final JSON output.

**Primary Functions**

- *post_filter(text: str, characters: list[str]) -> list[str]*: Applies dependency parsing to a sentence and returns the intersection between the grammatical subjects (*nsubj* or *nsubjpass*) and the LLM-generated character list.
- *filter_df(df: DataFrame) -> DataFrame*: Applies *post_filter* to each row in the LLM output DataFrame, processing the *coref_text* field and updating the *filtered_character* column accordingly.

Example of filtered output (*llm_character_dep* is the filtered character by dependency parsing):


```
{
    "llm_text": "As Gryffindors came spilling onto the field, he saw Snape land nearby, white-faced and tight-lipped — then Harry felt a hand on his shoulder and looked up into Dumbledore's smiling face.",
    "matched_gold_text": "As Gryffindors came spilling onto the field, he saw Snape land nearby, white-faced and tight-lipped — then Harry felt a hand on his shoulder and looked up into Dumbledore's smiling face.",
    "text_match_type": "exact_match",
    "llm_type": "narrative",
```

```
    "gold_type": "narrative",
    "llm_character": [
            "Dumbledore",
            "Harry",
            "Snape"
    ],
    "llm_character_dep": [
            "Harry",
            "Snape"
    ],
    "gold_character": [
            "Harry",
            "Snape"
    ]
}
```

#### 4.2.2.6 Output Data

The final output of the pipeline is a JSON file containing the following fields:

- *text*: Original sentence
- *coref_text*: Sentence with expanded coreference
- *type*: Sentence type (*narrative* or *direct_speech*)
- *character*: List of attributed characters
- *filtered_characters*: List of attributed characters after dependency parsing filter

Example of the LLM output:

```
{
    "book_id": 7,
    "text": "He was abruptly awake in the sour-smelling darkness; Nagini had released him.",
```

"coref_text": "Harry was abruptly awake in the sour-smelling darkness; Nagini had released Harry.",
"type": "narrative",
"character": [
"Bathilda",
"Harry"
],
"filtered_characters": [
"Harry"
]
},

## 4.3 Experiment 1: Comparative Study

This experiment benchmarks four large language models—**LLaMA-3-8b, Mistral-7b, GPT-4o, and Gemini 2.5 Pro**—on the task of extracting character sentences from continuous literary prose, with the goal of identifying the most suitable backbone for the proposed pipeline.

### 4.3.1 Experiment Design

**Models.** Two open-source models (LLaMA-3-8b, Mistral-7b) and two proprietary models (GPT-4o, Gemini 2.5 Pro) were selected. LLaMA-3-8b achieved >90 % quotation-attribution accuracy on the PDNC (Project Dialogism Novel Corpus) corpus in Michel et al. (2025), a task overlapping with character-sentence detection. Mistral-7b is widely adopted in literary NLP (Nešić et al., 2024; Siino, 2024). GPT models underpin previous character-focused studies (Dougherty, 2024; Jaipersaud et al., 2024; Pagel et al., 2024), and GPT-4o has recently excelled at span annotation for emotion categories (Jang and Jung, 2024). Gemini 2.5 Pro—Google's current flagship—has likewise been applied to character portrayal tasks (Jaipersaud et al., 2024).

As for the context length of the models, LLaMA-3-8b's context window is 8k tokens, Mistral-7b is 32k tokens, GPT-4o is 128k tokens, and Gemini 2.5 Pro has 1M token support for the input and 64k support for the output.

**Data.** Evaluation used a contiguous 200-sentence excerpt from *Harry Potter and the Philosopher's Stone* (segmented with BookNLP), coinciding with the manually annotated HPCS_full-sentence gold standard. The reduced sample keeps inference practical: a single LLaMA-3-8b run already requires 24 min 16 s, implying that processing the full gold corpus would exceed two hours.

**Process.** Each model received the cleaned, sentence-delimited passage produced by the pipeline's Text-Cleaning and Senticizer modules. No downstream dependency filtering was applied, so raw model outputs were assessed directly.

**Quality metrics.** Three quality metrics are tested for the LLMs' results:

- **Text identification metrics:** These metrics evaluate the pipeline's ability to correctly identify character sentences from coherent input texts. A prediction is matched to a gold sentence when Python's *difflib.SequenceMatcher* yields ≥0.85 similarity.
  - **Precision** = $\frac{text\ matched\ LLM\ entries}{total\ LLM\ entries}$
  - **Recall** = $\frac{text\ matched\ Gold\ entries}{total\ Gold\ entries}$
  - **F1** = $2 \times \frac{precision \cdot recall}{precision + recall}$
- **Character attribution metrics:** These metrics quantify how well a model assigns the correct character set to each sentence it has identified. We distinguish two match criteria—**Exact** (predicted set = gold set) and **Soft** (predicted set ⊆ gold set)—and compute each on

two scopes: **matched** – only the sentences that align with the gold corpus; **total** – all sentences produced by the model.

– $Accuracy_{to\_matched} = \frac{exact\ character\ match\ LLM\ entries}{text\ matched\ LLM\ entries}$

– $Accuracy_{to\_total} = \frac{exact\ character\ match\ LLM\ entries}{total\ LLM\ entries}$

– $Soft\ Accuracy_{to\_matched} = \frac{soft\ character\ match\ LLM\ entries}{text\ matched\ LLM\ entries}$

– $Soft\ Accuracy_{to\_total} = \frac{soft\ character\ match\ LLM\ entries}{total\ LLM\ entries}$

- **Type classification metrics:** These measure whether the sentence type is correctly classified as either *narrative* or *direct_speech*. **Accuracy** is reported.

– $Accuracy_{to\_matched} = \frac{type\ correct\ LLM\ entries}{text\ matched\ LLM\ entries}$

– $Accuracy_{to\_total} = \frac{type\ correct\ LLM\ entries}{total\ LLM\ entries}$

### 4.3.2 Results

Results from the comparative study are listed in Table 4.2. The total Gold text entries for the tested sample are 204. Results show that GPT-4o reaches the highest text identification accuracy, while the highest recall and precision are with Gemini 2.5 Pro. Gemini 2.5 Pro also achieves the best performance in the character attribution task overall. All LLMs except LLaMA-3-8b reach 100% accuracy in type classification within the text matched LLM entries.

**Table 4.2 Comparative study metrics**

| | | LLaMA-3-8b | Mistral-7b | GPT-4o | Gemini 2.5 Pro |
|---|---|---|---|---|---|
| Text identification | **LLM text entries** | 124 | **448** | 141 | 202 |
| | **Precision** | 99.19% | 85.94% | **99.29%** | 91.96% |
| | **Recall** | 59.31% | 34.31% | 67.65% | **99.51%** |
| | **F1** | 74.24% | 49.04% | 80.47% | **95.59%** |
| **Character attribution** | **Accuracy to matched** | 69.11% | 64.42% | 73.57% | **89.81%** |
| | **Accuracy to total** | 68.55% | 55.36% | 73.05% | **82.59%** |
| | **Soft accuracy to matched** | 70.73% | 66.23% | 75.00% | **91.75%** |
| | **Soft accuracy to total** | 70.16% | 56.92% | 74.47% | **84.38%** |
| **Type classification** | **Accuracy to matched** | 78.86% | **100%** | **100%** | **100%** |
| | **Accuracy to total** | 78.23% | 85.94% | **99.29%** | 91.96% |

### 4.3.3 Discussion

Across the text-identification task, all four models achieved high precision, with LLaMA-3-8b and GPT-4o exceeding 99%, confirming that large language models can accurately flag most character-related sentences when properly prompted. Yet substantial gaps emerged in recall: LLaMA-3-8b, Mistral-7b, and GPT-4o systematically missed many direct-speech lines—syntactically signalled by quotation marks—and therefore failed to recover the full set of portrayal sentences. Gemini 2.5 Pro alone balanced precision and recall, yielding the highest F1 score. Mistral's figures were distorted by hallucinated duplicates that inflated its output count. Besides, only GPT-4o and Gemini 2.5 Pro succeed in generating meaningful coreference expanded texts upon the prompt's instruction.

Performance diverged even more sharply on character attribution. Gemini again ranked first on every metric, whereas LLaMA-3-8b and GPT-4o frequently assigned sentences to every mentioned character rather than to the grammatical subject, contravening the study's attribution rule and depressing their exact-match scores; Mistral suffered the same duplication problem observed earlier. In sentence-type classification, the task's purely syntactic nature—presence or absence of quotation marks—produced near-ceiling results for Mistral, GPT-4o, and Gemini, with LLaMA-3-8b only marginally lower.

Taken together, these findings identify Gemini 2.5 Pro as the most robust model across all three evaluation dimensions, and it is therefore adopted as the core large language model in the subsequent pipeline implementation.

## 4.4 Experiment 2: Quality Evaluation

This experiment evaluates the performance of the proposed NLP pipeline on the full sentence-level Gold dataset. The goal is to assess the pipeline's effectiveness in three core tasks: identifying character sentences from full literary texts, attributing them to the correct characters, and classifying sentence types. This evaluation demonstrates the reliability of the pipeline in a realistic literary setting, which highlights the strengths and limitations of its current implementation.

### 4.4.1 Experiment Design

The pipeline is applied to the original *Harry Potter* text passages used to create the Gold dataset. The pipeline only takes one LLM, Gemini 2.5 Pro, which is proven to have the best performance on sentence identification and character attribution in the previous comparative studies (§4.3), as well as having the shortest run time.

The outputs are compared against *HPCS_full-sentence* (see §4.1) using three sets of metrics: Text identification metrics (precision, recall, F1), Character attribution metrics (accuracy, soft accuracy), and Type classification metrics (accuracy), see §4.3.1-Quality metrics for details. In this experiment, the pipeline activates the Dependency Filter module, which uses spaCy's dependency parser to discard LLM-attributed characters that appear solely as grammatical objects in the coreference-expanded sentences. Character-attribution accuracy is thus reported for two settings: LLM (raw LLM attribution) and LLM + Dependency (final output after filtering).

### 4.4.2 Results

Table 4.3 presents the pipeline's performance in identifying character sentences.

**Table 4.3 Text identification metrics**

| Metric | Value |
|---|---|
| Total Pipeline entries | 1429 |
| Total Gold entries | 1432 |
| Text match Precision | 95.24% |
| Text match Recall | 93.74% |
| Text match F1 | 94.51% |

Table 4.4 shows attribution accuracy with and without the dependency parsing filter.

**Table 4.4 Character attribution accuracy metrics**

| | To matched | | To total | |
|---|---|---|---|---|
| | **Accuracy** | **Soft accuracy** | **Accuracy** | **Soft accuracy** |
| **LLM** | 90.60% | 91.40% | 86.28% | 87.05% |
| **LLM + Dependency** | 89.13% | 93.75% | 84.88% | 89.29% |

Of the 1,429 total pipeline outputs, 1,359 sentences were classified with the correct sentence type. Type classification accuracy is 99.85% relative to matched texts and 95.10% overall. Table 4.5 reports accuracy by sentence type.

**Table 4.5 Character accuracy by sentence type**

| | Direct speech | | Narrative | |
|---|---|---|---|---|
| | **Accuracy** | **Soft accuracy** | **Accuracy** | **Soft accuracy** |
| **LLM** | 97.47% | 97.47% | 85.36% | 86.93% |
| **LLM + Dependency** | 97.47% | 97.47% | 82.88% | 91.11% |

### 4.4.3 Discussion

The pipeline achieves strong performance in identifying character sentences, with both precision and recall approaching 95%. Although no direct baseline exists, this result represents a high level of accuracy for an applied NLP task in computational literary studies (Bamman et al., 2019; Michel et al., 2025; Zhang and Liu, 2021).

Character attribution also performs well, with approximately 90% soft accuracy. The LLM alone shows strong attribution performance, while the dependency parsing filter helps reduce false positives. However, it occasionally removes correct attributions, slightly lowering strict accuracy. Manual analysis suggests that most false negatives occur when multiple characters share the grammatical subject role in a single sentence. For example, the sentence "They said" may expand via coreference to "Harry, Ron, and Hermione said," but dependency parsing may only identify "Harry" as the subject, leading to the exclusion of Ron and Hermione.

Attribution by sentence type shows particularly high performance for direct speech, a task closely related to quotation attribution. The current pipeline's results in direct speech attribution outperform Michel et al. (2025), who used LLaMA-3 8b with in-context learning.

Narrative sentences remain more challenging due to complex structures, shifting perspectives, and coreference ambiguities. Implicit quotations and free indirect speech, which lack explicit markers, further increase attribution difficulty. Despite this, Gemini maintains over 85% attribution accuracy for narrative sentences, demonstrating its robust contextual understanding.

## 4.5 Experiment 3: Prompt Optimization and Selection

This experiment evaluates different prompting strategies for the pipeline. The goal of this experiment is to optimize the prompt design and achieve the best pipeline performance. Four prompts have been tested, and three quality metrics (text identification, character attribution, and type identification metrics) are reported for each prompt. This experiment illustrates how prompt structure influences LLM performance in complex literary analysis tasks.

### 4.5.1 Experiment Design

**Prompt configurations.** This experiment evaluates four distinct prompt configurations employing advanced prompting techniques: **Chain-of-Thought (CoT)**, **LangGPT prompting framework**, and **two-step coreference resolution**. These prompts are tested at two annotation granularities: clause-level and full sentence-level character sentence identification. Only Prompt 4 is tested with the Dependency Filter module. Detailed prompt configurations and corresponding techniques are presented in Table 4.6 below.

**Table 4.6 Prompt configurations**

| | **Chain-of-Thought** | **LangGPT template** | **Two-step coreference reasoning** | **Clause-level** | **Full sentence-level** | **All Gold data** | **Book 1 Gold data** | **Dependency filter module** |
|---|---|---|---|---|---|---|---|---|
| **Prompt 1 (Base prompt)** | | | | ✅ | | | ✅ | |
| **Prompt 2** | ✅ | | | ✅ | | | ✅ | |
| **Prompt 3** | | ✅ | | | ✅ | ✅ | | |
| **Prompt 4** | | ✅ | ✅ | | ✅ | ✅ | | ✅ |

**Prompt 1**, the base prompt, does not include any prompting techniques or examples. The prompt was adapted from Michel et al. (2025)'s quotation identification prompt.

Chain-of-thought (CoT) is a series of step-by-step reasoning integrated into the prompt to guide the LLMs to perform complex reasoning, which improves the model's performance on multiple tasks (Wei et al., 2023). In **Prompt 2**, a zero-shot CoT is utilized with providing a CoT cue of "Please think step-by-step," and detailing six reasoning steps: sentence boundary determination, sentence categorization, clause splitting, attribution rules, duplication avoidance, and output formatting.

LangGPT prompting framework is a dual-layer prompt design framework that assigns a Role (i.e., literary expert) and task instructions (i.e., identification and attribution rules) to fulfill Goals (i.e., extract character sentences; see Wang et al., 2024). Please see Section 4.2.2's LLM Processing module for detailed introduction of the prompt design. **Prompt 3** first utilized LangGPT framework, and building on which, **Prompt 4** added a two-step coreference resolution reasoning to the prompt. The two-step coreference reasoning is inserted in the <*Rules*> section of the prompt, asking the LLMs to first replace any pronoun references into names in a sentence and save a *coref_text* field in the output, and then attribute the sentence based on *coref_text*. The purpose of the two-step coreference resolution is to enhance attribution precision by reducing the pronoun ambiguity, given LLMs' outstanding performance in coreference resolution (Hicke and Mimno, 2024; Le and Ritter, 2023). Notably, Prompt 4 is the final prompt selected for the pipeline.

**Data.** Since Prompts 1 and 2 are tested at the beginning stage of the project, they are initially only tested on the *HPCS_clause-level* dataset and only on *Harry Potter* Book 1's Gold data; yet, they have already caused poor results and massive hallucinations on the subset (see Table 4.7 below). As the base prompt and CoT are not going to be the final prompt for the pipeline, this study terminated the testing and did not continue to test Prompts 1 and 2 on the full Gold dataset. Prompts 3 and 4 are tested on the full *HPCS_full-sentence* Gold dataset.

**LLM.** All prompts are only performed with Gemini 2.5 Pro in this experiment because it has the best performance in the previous comparative studies, see §4.3.

**Metrics.** Three metrics are assessed for each prompt: Text identification metrics (precision, recall, F1), Character attribution metrics (accuracy, soft accuracy), and Type classification metrics (accuracy). Calculation of the metrics is the same as Experiment 1, please see §4.3.1.

Since Prompt 4 is assessed with the Dependency Filter module, Prompt 4's character attribution accuracy is tested on two settings: LLM (LLM's attribution) and LLM+Dependency (LLM's attribution after the dependency filter). Accuracy (exact match) and soft accuracy (*pipeline character* $\subseteq$ *Gold character*) are included in the metrics.

### 4.5.2 Results

The identification precision and F1 scores progressively improved from Prompt 1 to Prompt 4, although Prompt 3 achieved the highest recall (94.69%). Regarding character attribution (LLM-only), Prompt 4 yielded the highest strict accuracy (86.28%), while Prompt 3 demonstrated the highest soft accuracy (91.19%). Integrating the Dependency Filter module with Prompt 4 slightly reduced strict accuracy (from 86.28% to 84.88%) but improved soft accuracy (from 87.05% to 89.29%), though still below Prompt 3's soft accuracy by 1.9 percentage points. Type identification accuracy consistently increased, reaching 95.10% in Prompt 4. The detailed metrics for all four prompts are presented in Table 4.7.

**Table 4.7 Quality metrics for four prompts**

| | | **Prompt 1 Base prompt** | **Prompt2 CoT – clause-level** | **Prompt 3 LangGPT – full sentence** | **Prompt 4 LangGPT+two-step coref – full sentence** |
|---|---|---|---|---|---|
| | **Testing dataset** | Gold data – Clause-level – Book 1 | Gold data – Clause-level – Book 1 | Gold data – Full sentence | Gold data – Full sentence |
| | **Gold data entries** | 214 | 214 | 1432 | 1432 |
| | **Total pipeline entries** | 272 | 235 | 1521 | 1429 |
| **Identification** | **Precision** | 77.57% | 86.38% | 93.75% | **95.24%** |
| | **Recall** | 91.12% | 92.06% | **94.69%** | 93.78% |
| | **F1** | 83.80% | 89.13% | 94.22% | **94.51%** |
| **Attribution (LLM)** | **Accuracy to matched** | 86.26% | 88.18% | 89.62% | **90.60%** |
| | **Accuracy to total** | 66.91% | 76.17% | 84.02% | **86.28%** |
| | **Soft accuracy to matched** | 96.68% | 97.04% | **97.27%** | 91.40% |
| | **Soft accuracy to total** | 75.00% | 85.83% | **91.19%** | 87.05% |
| **Attribution (LLM + Dependency)** | **Accuracy to matched** | | | | 89.13% |
| | **Accuracy to total** | | | | 84.88% |
| | **Soft accuracy to matched** | | | | 93.75% |
| | **Soft accuracy to total** | | | | **89.29%** |
| **Type classification** | **Accuracy to matched** | 86.73% | 99.51% | **99.93%** | 99.85% |
| | **Accuracy to total** | 67.28% | 85.96% | 93.69% | **95.10%** |

### 4.5.3 Discussion

The results illustrate the effectiveness of the LangGPT framework in guiding analytical tasks for LLMs. Specifically, Prompt 3 (LangGPT) enhanced text identification precision by

7.37 percentage points compared to Prompt 2 (Chain-of-Thought). However, improvements in character attribution accuracy were less pronounced; the accuracy of character attribution relative to matched texts increased by only 1.44 percentage points between Prompts 2 and 3. This smaller improvement suggests that Prompt 3's increased overall attribution accuracy primarily resulted from better text identification. Nevertheless, differences in annotation granularity between Prompt 2 (clause-level) and Prompt 3 (full sentence-level) may have influenced these results.

The transition to full sentence-level granularity was motivated by manual inspections of Prompts 1 and 2, which revealed the LLM’s tendency to over-segment sentences into subject-verb clauses (including dependent clauses), conflicting with the defined scope of character sentences. Implementing the Senticizer module and instructing attribution at the full-sentence level effectively reduced over-segmentation and hallucination, consequently improving identification accuracy. Future pipeline enhancements may include adding a specialized module for handling compound sentences to enable accurate clause-level attribution.

Prompt 4, incorporating the two-step coreference resolution, improved character attribution strict accuracy by 2.26 percentage points compared to Prompt 3 (LLM-only). Additional analyses indicated that, without coreference expansion, the LLM predominantly attributed only a single character per sentence, while coreference resolution enabled accurate attribution to multiple characters. However, this adjustment also introduced false positives, reducing soft accuracy by 4.14 percentage points. Addressing this trade-off will require further refinement of the attribution mechanisms.

Prompt 4 is selected as the optimized prompt for the pipeline as it has the capacity to handle multi-character sentences while maintaining a high F1 score in the character sentence identification and a high accuracy in character attribution.

### 4.6 Conclusion

This chapter first presented *HPCS*, two gold-standard datasets for character-sentence research, and *Ext-cs*, a complete LLM-driven pipeline that combines rule-based preprocessing, large language model reasoning, and dependency filtering.

Three experiments evaluated the pipeline in this chapter. **Experiment 1** compared four state-of-the-art LLMs for the pipeline's LLM Processing module, optimizing Gemini 2.5 Pro as the model. **Experiment 2** confirmed that the system detects character sentences with ≈ 95 % precision and recall and achieves ≈ 90 % soft attribution accuracy. **Experiment 3** highlighted the impact of prompt engineering: the LangGPT template sharply boosted sentence identification, while adding two-step coreference reasoning raised strict attribution accuracy at the cost of a slight drop in soft accuracy. Prompt 4 is adopted as the pipeline's optimized configuration because it handles multi-character sentences well and maintains strong performance across all other metrics.

The experiment results confirm that API-accessible LLMs can enable fine-grained literary analysis at scale. Future work will focus on handling compound sentences for clause-level attribution, integrating semantic-role labeling to refine multi-character attributions, and benchmarking the pipeline on complete novels and diverse corpora. The datasets, code, and prompts released with this thesis provide a reproducible springboard for subsequent research in computational literary studies.

# Chapter 5: Conclusion

## 5.1 Summary of Research

### 5.1.1 Objectives and Major Outputs

This research aims to develop a theory-informed computational method to study literary characterization. The objectives include: (1) defining and formalizing the analytical unit of "character sentence" (CS), (2) creating Gold datasets and an NLP pipeline integrated with large language models (LLMs) for CS extraction and benchmarking, (3) optimizing model selection and prompt design, and (4) open-sourcing the project for future research.

Three key outputs are achieved:

First, the concept of a character sentence is defined along with clear operational criteria. A character sentence includes any literary sentence conveying portrayal details about recognizable characters, encompassing descriptions (attributes, relations, emotions) and actions (dialogue, physical actions, mental processes). A complete quoted dialogue is treated as a single CS, attributed to its speaker, simplifying computational extraction and enhancing accuracy.

Second, two annotated Gold datasets (HPCS_clause-level and HPCS_full-sentence) are created for benchmarking automated CS extraction. These datasets, sampled from the *Harry Potter* series, each contain 200-sentence passages annotated according to the defined criteria.

Third, the NLP pipeline, Ext-cs, is developed and evaluated. Ext-cs comprises four modules—text cleaning, senticizer, LLM processing, and dependency filter. The pipeline segments and cleans input texts, identifies and attributes CS using Gemini 2.5 Pro with optimized prompts, and refines results through dependency parsing to ensure accurate character attributions. Comprehensive experiments assessed model optimization, pipeline performance, and prompt strategies. The pipeline, datasets, and code are publicly available under an open license.

### 5.1.2 Key Research Findings

One key finding is the conceptual validity of the defined character sentence. The operational criteria yielded high inter-annotator agreement (0.81 and 0.86) on the Gold datasets, confirming the robustness and reproducibility of the definition. The pipeline achieved strong performance, with an identification F1 of 94.51% and attribution accuracy of 84.88%, validating the sentence-level computational capture of character portrayal.

Another key finding highlights Gemini 2.5 Pro's robustness in computational character studies. Comparative evaluations demonstrated its superior recall and F1 scores compared to LLaMA-3-8b, Mistral-7b, and GPT-4o, marking it as highly effective and accessible for future computational literary analyses.

The research also identifies effective prompting techniques, specifically the LangGPT framework combined with two-step coreference resolution. This structured, stepwise approach significantly improved LLM accuracy over dominant methods such as zero-shot chain-of-thought (CoT) prompting. Two main reasons explain this improvement. Firstly, although both LangGPT and CoT provide step-by-step instructions, LangGPT offers a more structured format by clearly separating rules and constraints, aligning with recent findings in prompt engineering (Li et al., 2025). Secondly, the two-step coreference resolution breaks down the task into two clear subtasks: resolving coreferences first and then attributing characters based on the resolved text. This explicit division enhances stepwise reasoning for the LLM, effectively integrating coreference knowledge with the attribution task.

### 5.1.3 Answering Research Questions

Four research questions (see §1.4) guided this study:

The first question was how a CS can be defined such that it is both distinctive in narratology and computationally tractable. Cognitive narratology underscores that (implied) readers identify characterization cues in specific textual instances, implying that certain texts distinctly trigger characterization processes. In this research, CS are classified into descriptions and actions, aligning with narrative theories and creative writing practices (La Ronn, 2021; Lubbock, 1921; Murray, 1916). The operational definition and criteria were designed to consider NLP and LLM capabilities. Particularly, treating a complete quoted dialogue as one CS preserves essential contextual information, distinctly separating dialogues from narrative descriptions. This design choice simplifies the syntactical task for LLMs, such as clearly marking dialogue boundaries, thus enhancing accuracy in character sentence identification and attribution.

The second question evaluated to what extent an API-accessible LLM can accurately detect and attribute CS in complex literary prose. Experiments indicated Gemini 2.5 Pro's superior capability (see metrics in §5.1.2), particularly when optimized with structured prompting techniques.

The third question assessed the strengths and limitations of the Dependency Filter (rule-based NLP) module within the Ext-cs pipeline. This module effectively reduced false positive character attributions, increasing soft accuracy by 2.84%. However, strict accuracy decreased slightly (by 1.4%) due to the parsing model's limitation of recognizing only one grammatical subject per verb. For instance, in sentences like "Harry, Ron, and Hermione said," the parsing model identifies only the first named character as the subject, potentially removing correctly identified additional characters. Despite this limitation, reducing false positives significantly benefits downstream analyses by ensuring greater precision in character attributions, thereby supporting more accurate character modeling and interpretation.

The last research question inquires what prompting strategies maximize LLM performance. Ablation studies confirmed that integrating the structured LangGPT framework with explicit two-step coreference resolution significantly enhanced LLM performance, particularly for character identification and attribution tasks.

## 5.2 Implications

### 5.2.1 Implications for Cognitive Narratology

The consistently high inter-annotator agreement in the CS Gold datasets and the pipeline's strong extraction scores together indicate that the textual cues hypothesized by cognitive narratology are (a) reliably recognized by human annotators and (b) automatically detectable by state-of-the-art language models. This convergence lends empirical support—albeit indirect—to the core claim that readers draw on relatively stable linguistic signals when inferring fictional minds and emotions. Because the present study does not measure real-time reader cognition, it stops short of confirming how those inferences unfold; however, its results supply an encouraging foundation for follow-up work, such as experiments that ask participants to locate character sentences and compare their judgments with the Gold standard.

Within cognitive narratology, scholars disagree about the precise role of text in character construction. Culpeper argues for an interaction between bottom-up textual information and readers' top-down knowledge; Zunshine frames textual features as both triggers and manipulators that activate Theory-of-Mind (ToM) reasoning; by contrast, Pettersson's cluster conception treats the physical text merely as an interface, insisting that all “textual information” is mentally constructed (Tevdoradze, 2025; Zunshine, 2003, 2006). The present findings can inform this debate: regardless of whether textual cues are seen as information “given” or as affordances for inference, the implied readers appear to agree on which sentences pertain to characterization and which do not. Whether the actual readers engage in selective cognitive processing of those cues, and how such processing unfolds, remains an open empirical question.

In short, the character-sentence framework equips cognitive narratology with a reproducible toolset that can translate long-standing theoretical questions about readerly mind attribution into testable research programs.

### 5.2.2 Implications for NLP and CLS

One implication from the designed NLP pipeline is that sentence-level boundary marking stabilizes LLM extraction. Prior work that asked large language models to highlight character-related spans without predefined boundaries achieved at best 64.8 % span match (Jang and Jung 2024), largely because boundary mismatches dominated the error profile; similar degradation was reported by Ma et al. (2023) for classification tasks. In the present study, the Senticizer module supplies explicit sentence segmentation before prompting, and every model–prompt combination exceeds 80 % precision in text identification (see §4.3 and §4.5). These results confirm that boundary labelling offers a simple yet effective alignment mechanism for extraction pipelines.

Prompt-plus-parser hybrids rival fully supervised systems. A two-stage design—structured prompting (LangGPT with two-step coreference) followed by a lightweight dependency filter—pushes Gemini 2.5 Pro to an F1 of 94 % without any fine-tuning. The finding strengthens the case for treating prompt structure as a tunable hyperparameter and for combining generative LLMs with symbolic post-processing when high precision is required.

Literary scholars who have limited programming experience can nevertheless perform sophisticated character analyses with LLMs: extraction, profiling, sentiment, or action typology, etc. Recent studies (Jang and Jung , 2024; Michel et al. , 2025; Yuan et al. , 2024) corroborate that LLMs handle narrative tasks well. However, researchers must acquire a working knowledge of LLM behavior, such as API usage, structured output formats, prompt design, and hallucination control, and appreciate the supporting role of classic NLP modules such as sentence segmentation and dependency parsing. The pipeline presented here illustrates how those components can be orchestrated to deliver reproducible, high-precision results at scale.

## 5.3 Contributions

**Theoretical advancements.** This research introduces and rigorously defines character sentences as a narratology-grounded, sentence-level unit of characterization. The inter-annotator agreement and the performance of Ext-cs, the NLP pipeline for automated CS extraction, demonstrate that CS cues are consistently identifiable by humans and automatically detectable by state-of-the-art LLMs. The theoretical implications for cognitive narratology, as mentioned in 5.2.1, are encouraging further studies to use CS as materials to study the readers' Theory-of-Mind in reading. The theory-to-tool workflow of the research, from conceptualizing CS to developing the NLP pipeline, shows how characterization theories of "showing & telling" and action-centered poetics (Aristotle) map cleanly onto CS sub-types and extraction rules, giving narratology a computationally testable substrate.

**Methodological innovations.** This research publishes two openly licensed Gold datasets (HPCS_clause-level and HPCS_full-sentence) that benchmark the CS extraction task, which includes quotation detection and attribution, coreference resolution, and subject attribution tasks that were previously treated in isolation. This study also designs Ext-cs, the first LLM-integrated pipeline for CS identification and attribution. The LLM performance of the pipeline shows that a hybrid of structured prompting and symbolic post-processing (the Dependency Filter module) attains high performance without any model fine-tuning. This study is also the first to evaluate the LangGPT prompting framework in a literary context, establishing that it is a powerful, reusable prompting scaffold. The metrics used to evaluate the pipeline in this research (i.e., text alignment for CS identification, strict and soft accuracy for CS attribution) are also reusable for future work.

**Applied contribution to NLP & LLMs.** This study expands the suite of narrative information extraction benchmarks by adding a sentence-level characterization task, highlighting a new stress test for LLM narrative competence. This study also implies that explicit sentence segmentation or text boundary setting stabilizes generative information extraction with LLMs (Jang and Jung 2024). This study also tested Gemini's performance in information extraction in literary and humanities contexts, establishing a benchmark for its future utility.

**Applied contribution to CLS.** CS enables more accurate sentence-level downstream analyses, including emotion profiling, personality modeling, agency studies, etc. With characterization-focused CS, the downstream tasks can make sure that their modeling is target character-focused, avoiding the noisy results from analyzing the sentences that merely mention names. This study also elevates awareness of sentences as the analytical units in distant reading, complementing the existing word- or document-level approaches. This research encourages CLS researchers, especially non-coders, to adopt LLM-based pipelines by demonstrating a turnkey, prompt-driven workflow. Lastly, all code, data, and annotation guidelines are version-controlled and MIT-licensed, setting a transparency standard in CLS.

### 5.4 Limitations

One limitation of this research is that the analyses are conducted based on the implied readers, not the actual readers. This thesis uses LLMs to exemplify the implied readers' reading and characterization process by asking them to identify and attribute character sentences, yet there is no way in this research to examine the actual human cognition behind the process. However, previous empirical studies in cognitive narratology and reader-response theory show that actual readers' responses toward the text resemble the theoretical, implied readers. In this research, the pipeline's output also has high consistency with the annotated Gold dataset, showing that the implied readers' alignment with the actual readers. Future neuro-cognitive or psycholinguistic studies investigating how real readers cognitively engage with character sentences are necessary to address this gap fully.

Another methodological limitation is that the NLP pipeline evaluation is conducted only on relatively short, 200-sentence-length passages. Although the sliding window mechanism in the LLM Processing module is designed to handle longer texts, increasing input lengths might elevate the risk of hallucination and error rates. Future developments of the pipeline will need improved error-handling and hallucination mitigation strategies to manage longer and more complex narrative texts reliably.

The dataset itself presents limitations on the generalizability of the research. The Gold datasets used for benchmarking were exclusively sampled from the *Harry Potter* series. While these novels exemplify general genre fiction and have been widely analyzed in computational literary studies and NLP, their stylistic homogeneity—being written by a single author, J.K. Rowling—limits the dataset's variety. Consequently, findings based on these datasets may not fully generalize to contemporary literary texts that differ markedly in style or authorial approach. Future research should consider incorporating more diverse literary sources, such as the larger Project Gutenberg corpus, to improve the generalizability of these findings.

There are also limitations associated with the prompting strategy utilized in the pipeline. Specifically, the current two-step coreference expansion strategy requires the LLM to output each character sentence twice: once in its original form and once with resolved coreferences (the same sentence only with pronouns like he and she replaced by character names), leading to redundancy and token inefficiency. Using OpenAI's token counter, it was observed that for the sample from the first *Harry Potter* book, the output tokens (19,913) more than doubled the input tokens (9,187, counting the prompt and the senticized sample text together), indicating substantial inefficiency. Although this redundancy enables more accurate dependency parsing, future iterations should explore more compact output data structures to optimize token usage.

Finally, this study's reliance on the Gemini 2.5 Pro (free-tier) model imposes inherent technical limitations. The free-tier API limits (5 requests/minute, 25 requests/day, and a maximum of 1,000,000 tokens/day) significantly restrict processing larger literary corpora. While upgrading to paid tiers would mitigate these constraints, the use of a proprietary, closed-source model like Gemini 2.5 Pro runs counter to the open ethos promoted by Digital Humanities. Thus, despite the open availability of the datasets and code under the MIT license, the core dependency on a closed LLM service represents a notable limitation in terms of accessibility, transparency, and replicability.

## 5.5 Future Research

**Integration of CS in cognitive narratology research.** Future theoretical development in cognitive narratology would benefit from integrating CS into empirical studies. Prior empirical research exploring the connection between readers' interpretations of fictional characters and Theory of Mind (ToM) processes has relied on qualitative or loosely defined textual stimuli (Kidd et al., 2016; Zunshine, 2006). Character sentences, extracted systematically from literary texts, offer a more controlled and quantifiable stimulus set. For instance, future studies might conduct experiments in which one participant group reads texts rich in CS, while a control group reads texts with fewer CS. The participants' ToM could then be measured using assessments like the Reading the Mind in the Eyes Test (RMET; Baron-Cohen et al., 2001). Neuroscientific approaches, such as functional brain imaging (fMRI) or electroencephalography (EEG), could similarly leverage CS-rich texts to investigate brain activation patterns related to reading fictional narratives. Previous studies have categorized literary materials broadly as vivid versus abstract or social versus non-social without quantifiable textual standards (Tamir et al., 2016). Future research could use CS proportions and character interactions within CS as quantitative measures to categorize and select experimental texts more systematically.

**Applications of CS in downstream computational literary studies (CLS) tasks.** Character sentences provide a robust foundation for a variety of downstream CLS tasks:

- **Affect analysis of CS:** One potential direction involves affect analysis, encompassing sentiment polarity and categorical emotion analysis. By applying sentiment algorithms to CS, researchers can construct detailed emotional arcs depicting how a character's emotional state evolves throughout a narrative. Categorical emotion analysis could similarly classify each CS into discrete emotion categories (e.g., happiness, sadness, anger) and visualize these emotions with tools such as radar charts. Researchers might also identify emotion experiencers and triggers from CS to map emotional interaction networks among characters, providing nuanced insights into emotional complexity and character relationships.

- **Personality profiling with CS:** Another promising direction involves personality profiling using psychological frameworks such as the Big Five (OCEAN) personality model. Applying personality prediction algorithms to a character's set of CS can reveal both overall personality tendencies and detailed personality trait distributions. Such personality modeling would closely reflect explicit textual portrayals, offering computationally derived character profiles consistent with narrative characterization.
- **Visualizing character interactions:** Additionally, analyzing CS co-occurrences across texts enables visualization of character interactions, beneficial for creative writers and content creators to evaluate narrative alignment with intended character dynamics and distributions.

Future research could apply the current pipeline to examine differences in characterization between fanfiction and its canon (the original work on which the fanfiction is based). In fanfiction writing, some authors strive to preserve the canonical characterization faithfully, labeling deviations as "OOC" (out of character), while others intentionally alter characterization to express dissatisfaction with, or even resistance to, the source material. This phenomenon reflects the subtle dynamics, sometimes even a power structure, between the "official" version of a cultural work and its fan community. Character sentences (CS) can serve as a means of quantifying and investigating these nuanced relationships. By extracting CS from both fanfiction and the corresponding canonical work and conducting downstream quantitative analyses (such as character profiling), similarities and differences in characterization between the two can be revealed. Although this project originally intended to undertake such an analysis, it has not yet been carried out; in future work, I aim to apply CS to this comparative scenario.

**Expanding pipeline application to broader literary contexts.** Addressing the dataset limitations discussed earlier (§5.4), future research should test and refine the Ext-cs pipeline across a wider literary context. This includes contemporary fiction, online literature, fanfiction, poetry, and dramatic texts. For instance, contemporary authors such as Sally Rooney frequently employ implicit dialogue structures without quotation marks. Future pipeline enhancements should incorporate implicit quote detection mechanisms within the Senticizer module, accompanied by tailored rules in the LLM processing module. Likewise, diversifying the Gold datasets to include various literary genres and styles can substantially improve the pipeline's generalizability. For poetry and drama, whose textual structures differ notably from prose fiction, dedicated fine-tuning and methodological adaptations may be required.

**Methodological enhancements.** Several methodological improvements can further enhance the pipeline's robustness and efficiency:

- **Assessment on full-length literary works:** Currently, pipeline evaluation is limited to 200-sentence excerpts. Future research should test pipeline performance comprehensively on complete literary works, assessing the handling of extended contexts and evaluating long-range dependencies and hallucination risks in detail.
- **Sparse output data structures:** Future iterations might adopt sparse data structures, such as explicit coreference chain annotations (Hicke and Mimno, 2024), to achieve greater token efficiency while maintaining accuracy.

- **Exploration of open-source LLMs:** While the study primarily utilized Gemini 2.5 Pro due to performance considerations, this proprietary model's limitations conflict with open digital humanities values. Future research should explore more extensively the viability of open-source LLM alternatives, such as LLaMA and Mistral, identifying effective strategies to optimize their performance to levels comparable to closed-source counterparts.

## 5.6 Conclusion

This thesis introduced the character sentence (CS) as a sentence-level, theory-grounded unit for analyzing textual evidence of characterization. Anchored in cognitive narratology and reader-response theory, the concept assumes that implied readers infer stable traits from specific linguistic cues; the CS definition translates those cues into a form that both human annotators and large language models (LLMs) can reliably recognize.

To operationalize this idea, the study produced two open Gold datasets sampled from the *Harry Potter* series and Ext-cs, a four-stage pipeline that combines structured prompting with symbolic post-processing. These resources make characterization—a domain long associated with close reading—amenable to large-scale, replicable analysis. The extracted character sentences underpin downstream tasks such as emotion arcs, personality profiling, and interaction network visualization.

Although the present work does not measure reader cognition directly, it supplies theory-informed stimuli that future psycholinguistic and neuro-cognitive studies can exploit. More broadly, it demonstrates how state-of-the-art LLMs and prompt-engineering strategies can extend, rather than replace, interpretive practice within digital humanities.

In sum, this thesis offers both a conceptual lens and a methodological toolkit for studying how narrative prose breathes life into fictional figures. By aligning narratological theory with computational method, it establishes a foundation on which subsequent research can test richer hypotheses about characters, stories, and the minds that read them.

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# Appendix: Full Prompt of Ext-cs’s LLM Processing Module

````
# Role
Literature expert

## Profile
- Language: English
- Expertise: Sentence-level characterization & character-building understanding
- Output format: **JSON object** with no extra text, e.g.:
  ```json
  {
    "text": "Sentence from the text",
    "coref_text": "Sentence after coreference resolution",
    "type": "direct_speech | narrative",
    "character": ["Name1", "Name2", "..."],
  }

## Goals:

Identify characterization-related sentences from the given text and attribute them to their portrayed characters.

## Rules:

- **Characterization-related:** A sentence is considered characterization-related if it shows or implies a character’s action, thought, dialogue, emotion, or attribute.
  - If a sentence mentions or implicitly refers to at least one character (including pronouns like "he", "she", "they") or direct speech, you must include it. Otherwise, exclude.
- **Sentence Boundaries:** Each sentence is pre-marked (e.g., `|1| ... |1|`). Do not split or combine them. Omit the sentence ID markers in the output.
- **Types:** If the entire sentence is within quotes, use `"type": "direct_speech"`; If the sentence is not fully quoted, set `"type": "narrative"`.

### Rules for direct speech:

- **Attribution:** Resolve the speaker by context and attribute the sentence only to the speaker(s).
- `coref_text` = `text` (no pronoun expansion)
````
````

- `"character"` must contain only one name except when multiple characters speaking together.

### Rules for narrative sentences:

- **Two-stage coreference reasoning:**
  1. **Coref_expand:** Create `coref_text` by replacing *pronouns* with their antecedent names.
  2. **Who-acts filter:** Scan `coref_text`. Identify the *grammatical subject* (doer / experiencer) in the sentence. **Only subjects go into the `"character"` list.** Ignore characters that appear **only** as objects, possessives, or inside a quote that someone else speaks.
- **Reporting Clause**: If it's something like "She said, ..." attribute it initially to the speaker.
- **Single or Multiple Actors**: If a sentence includes multiple characters each acting or reacting, include them all in `"character"`.
- **Pronoun Resolution**: Resolve he, she, they, etc. to the correct names; if genuinely unclear, use `"Unknown"`.
- **POV vs. Action**: Attribute actions to the doer, not to the point-of-view holder.
- **Free-Indirect Speech**: If a sentence reflects one character's inner voice, attribute only to that character.

### Global constraints
- **Direct speech** → exactly **one** name in `"character"` (the speaker).
- If no clear subject, output `"Unknown"`.

## Workflow

For each sentence (pre-marked with `|id| ... |id|`):
1. Check if the sentence is characterization-related.
2. Determin `direct_speech` or `narrative`.
   1. If the sentence type is `direct_speech`, apply <Rules for direct speech>.
   2. If the sentence type is `narrative`, apply <Rules for narrative sentences>. Expand coreference and use `coref_text` to attribute characters to the sentence.
3. Use `coref_text` to attribute character(s); apply the above attribution rules.
4. Return only the JSON object. No extra words.

## Initialization:

As a <Role>, your goal is to <Goals>. You must follow the <Rules> and <Workflow>. No extra output beyond the final JSON.

## Character dictionary


```
{
  "Harry": {
    "names": [
      "Harry",
      "Potter",
      "Harry Potter",
      "Mr. Potter",
      "Mr. Harry Potter",
      "HARRY POTTER",
      "POTTER",
      "the famous Harry Potter",
      "Mr. H. Potter"
    ],
…
}
```